Nearly half of junction-spanning gapmer designs against the *NR4A3* fusions of extraskeletal myxoid chondrosarcoma pair a wild-type parent gene, and a longer catalytic gap cannot separate them

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Abstract

Extraskeletal myxoid chondrosarcoma is an ultra-rare sarcoma in which a variable partner gene fuses to *NR4A3*. The fusion transcript carries a short sequence spanning the breakpoint junction that occurs in no normal gene. An antisense drug could in principle destroy that transcript by base-pairing alone, sparing both genes the fusion was built from, but none has been reported for this disease. Designs prove easy to find. Of the 231 ways the five known partners could join *NR4A3*, 38 keep the reading frame intact, and all 38 yield candidate gapmers, whose central DNA gap is where RNase-H1 cuts. Sparing the parent genes is the established test of such a design, and the sequence screens that rank candidates beforehand cannot see the form the liability takes: an identity-filtered alignment screen cannot surface an eleven-base-pair contiguous duplex, and a mature-transcript screen cannot reach precursor RNA. Compared directly against the six parent transcripts, 87 of 190 candidates pair that gap against a mature parent transcript over a contiguous duplex of at least ten base pairs, 61 of those against healthy *NR4A3*, the gene the approach exists to spare. That rate is not what arbitrary sequence gives: scrambled and randomly chosen 16-mers reach 6.2% and 6.9% on the same screen, and chimeras joining the same two parent transcripts at random offsets reach 23.8%, against 45.8% observed. Another 19 candidates pair the gap in unspliced precursor RNA, where RNase-H1 is also active and no mature-transcript screen reaches. Lengthening the gap, the obvious remedy, cannot help, and the reason is arithmetic rather than empirical: in every design of all three geometries tiled here, the junction-unique bases a longer gap wins and the contiguous wild-type-parent duplex it concedes are the same nucleotides. Candidates do clear the parent screens at every junction with a published exon-resolved breakpoint, and nine junctions now hold both such a breakpoint and a design carried through all five screens — four of them at acceptors the frame grading excludes and an RNase-H mechanism does not. The two most frequently reported of the nine are together 68.4% of molecularly confirmed cases, roughly two thirds, once partner prevalence is discounted by the breakpoint distribution of a single series, on which the whole set is 79.0%; discounted instead by a distribution pooled across two series the same two are 67.1% and the whole set is 82.9% (57.5–90.7%). They do not clear the screens cleanly: the *EWSR1* exon-12 reagent named below carries the heavier of the two reagents' disclosed transcriptome loads, 123 gap-paired sense-strand near-matches at six gene loci, together with a sense-strand near-match in wild-type *TAF15* precursor RNA. A liability no parent screen can reach is reported here as well. Where a design's acceptor half is *NR4A3* intronic or 5′ untranslated sequence, the same sequence sits immediately downstream of an intron in the patient's un-rearranged *NR4A3* allele, and three designs that had each passed the spliced-cDNA parent screen pair their whole catalytic gap there. Which designs do so is decided by how much donor sequence the gap holds, not by the gap-level margin they are ranked on. Two reagents are named for synthesis with those loads attached — 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6 — with those three named as not to be used, alongside the controls that make a knockdown

experiment interpretable and the selectivity value that would falsify the ranking used here. The work is computational: no wet-lab experiment was performed, no sequence named has been synthesised or tested, and nothing here asserts efficacy, safety, delivery to a tumour or clinical readiness.

1 · Introduction

EMC is defined in the large majority of cases by an in-frame fusion of *EWSR1* to the orphan nuclear receptor *NR4A3*,[1] with *TAF15* accounting for a substantial minority and *TCF12* and *TFG* reported rarely.[2] *FUS::NR4A3* is reported in a recent series that identified it by sequencing in two of five variant EMCs.[3] Next-generation sequencing of six EMCs finds few recurrent secondary mutations beyond the fusion,[4] so it is to a first approximation the single clonal driver. In every junction type described, the predicted product joins the amino-terminal transactivation domain of the partner to essentially the entire NR4A3 protein, including its nuclear-receptor DNA-binding domain.[1,5]

That driver is currently untargeted. Surgery with clear margins is the backbone of localised disease, and for advanced disease no clinically validated agent directly targets *NR4A3*.[6] The largest EMC-specific prospective study, a single-arm phase 2 of pazopanib in centrally confirmed *NR4A3*-translocated disease, returned four objective responses in 22 evaluable patients.[7] Anthracycline-based chemotherapy returned four responses in ten evaluable patients in a molecularly confirmed retrospective series of eleven, a result that series presents as running counter to the prior record.[8] The two sources cited above report a low objective response rate to that chemotherapy and low sensitivity to cytotoxic chemotherapy generally,[6,7] and none of the three publishes a response rate by line of therapy. The population a fusion-directed agent would address is close to the whole disease: across 58 molecularly confirmed cases, 79% carried *EWSR1::NR4A3*, 16% *TAF15::NR4A3* and 3% *TCF12::NR4A3*.[9]

The chimeric mRNA offers a discrimination handle its protein product does not. The *NR4A3* ligand-binding domain is retained near-intact in the chimera and is identical in sequence to wild-type NR4A3, so a ligand that engages it cannot distinguish fusion from wild type. The breakpoint junction can: it is a contiguous stretch of sequence present in no normal transcript, and absent from both parent transcripts, so selectivity can in principle be enforced by base-pairing rather than by protein conformation. That junction is the target of every design in this paper.

Targeting a fusion breakpoint with an oligonucleotide is not new. The approach has a continuous lineage from 1991,[10] including RNase-H-dependent antisense at a sarcoma fusion breakpoint in 1997,[11] and the fusion-exclusivity rationale was stated as a general principle in 2005.[12] Parental sparing has been reported at four fusions, two of the four with a readout on the wild-type parent transcript and the other two asserting junction specificity without one.[13–16] A bi-shRNA lipoplex directed at the *EWSR1::FLI1* junction was taken to preclinical justification,[17] and a GalNAc-conjugated junction siRNA in fibrolamellar hepatocellular carcinoma passed the delivery gate in a rare fusion-driven cancer.[18] The contribution here is therefore not the modality but the indication. Across 5,153 unique records retrieved from Europe PMC, four mention *EWSR1::NR4A3* at title or abstract level, resolving to three papers, none an oligonucleotide study.

Two questions follow that the field has not asked of this disease. The first is that prior design work has addressed only *EWSR1*, while the partner varies. Partner identity may not be clinically inert: across the two series of antiangiogenic tyrosine-kinase inhibition in advanced EMC that report a partner breakdown at all, no objective response is reported in a *TAF15* patient, on a *TAF15* arm of three to five patients whose Wilson upper bound remains compatible with equal response.[7,19] Neither primary report states the per-arm denominators; the sunitinib report states the partner split only qualitatively, and both denominators are read from published reviews of the two series.[20,21] The second is whether a junction oligonucleotide must be bespoke per patient, or whether one sequence can serve more than one fusion. That determines whether the deployable artefact for an ultra-rare disease is a stock reagent or a panel.

This paper answers both questions from sequence, sets out where a computational screen stops being able to answer them, and ends by naming the oligonucleotides to synthesise, the controls that make a knockdown result interpretable, and the pre-registrable threshold that would falsify the ranking every candidate here is ordered by (§5).

2 · Methods

Transcript models. Canonical transcripts for the five partner genes and for *NR4A3* were obtained from Ensembl.[22] Each model was self-checked before use: exon lengths must sum to the spliced cDNA, the CDS must occur exactly once within it, and translation of the CDS must reproduce the annotated protein. Per-exon coding content was additionally cross-checked against an independent exon audit for *EWSR1* and *NR4A3*; for the other four partners that audit does not exist, and the weaker check is recorded per gene in the released artefacts. Every exon number, coordinate and length in this paper is relative to one specific model per gene, and the canonical transcript of a gene can change between Ensembl releases, so the six accessions are given here rather than left to the artefacts: ENST00000397938 (*EWSR1*), ENST00000605844 (*TAF15*), ENST00000333725 (*TCF12*), ENST00000254108 (*FUS*), ENST00000240851 (*TFG*) and ENST00000395097 (*NR4A3*).

Chimera construction. Chimeras were built from transcript sequence rather than by joining coding sequences. A fusion keeps the whole *NR4A3* acceptor exon, so any bases of that exon lying ahead of the *NR4A3* start codon are still present in the fusion transcript, and they are the first bases an oligonucleotide meets on the *NR4A3* side of the junction. At the exon-3 acceptor, the only one that yields designs here, there are two. Joining coding sequences alone would omit them, shifting every design by two positions. A pair of exons is *in frame* when the partner's coding bases, plus those retained bases, sum to a multiple of three. Every declared exon pair was graded by that rule before any design was emitted, and only the in-frame pairs were carried forward, since only those describe a fusion that could exist.

Design. Junction-spanning 16-mer gapmers were tiled in a 5-6-5 LNA/DNA/LNA architecture on a phosphorothioate backbone, which is the chemistry the design rules below assume. Each way of sliding that 16-mer along the transcript is a *register*, and only registers placing the junction inside the six-nucleotide DNA gap were retained, since RNase-H1 cleaves within the DNA:RNA duplex of the gap and needs a minimum run of contiguous DNA to do so.

The gap length is a compromise and is treated as one. Reported minima for that run are five to six nucleotides: one states that RNase-H1 requires a gap of at least five for cleavage to occur,[23] and the other is a design-protocol statement that the DNA segment must be "six or more bases" to activate the enzyme.[24] Neither is a titration in this architecture, and in each the figure is given in passing, citing prior work rather than measuring it. For LNA/DNA/LNA gapmers specifically a six-nucleotide gap gives noteworthy but incomplete activity, with seven to ten reported as optimal; both figures are taken from a review that credits them to named earlier primary studies rather than measuring them itself.[25] Six therefore sits at the short end of the usable range and below the reported optimum. It was retained because it admits exactly five junction-spanning registers per junction. No claim is made that a short gap improves fusion-versus-parent discrimination: one series that shortened a 5-10-5 gapmer to 5-6-5 reported lower off-target knockdown but also lower on-target activity and lower allele selectivity.[23] Because that trade is the modality's central one, two longer geometries were tiled over the same junctions by the same rule and carried through the same screens: 5-8-5 and 5-10-5, with the wings held at five nucleotides so that only the gap changed. Holding the wings fixed is what makes the geometries comparable, since LNA affinity then enters every parent duplex identically. §3.8 reports the comparison.

Ranking. What separates the fusion from a parent transcript is the junction-unique bases inside the gap, not identity across the whole oligonucleotide, because the gap is where the enzyme cuts. Designs were therefore ranked by their *gap-level margin*: the number of junction-unique bases inside the gap on the shorter side of the junction. That is the ranking statistic for the panel-level results, which compare designs across junctions. Selecting within one junction is a different question and Table 4 answers it differently: designs there are ordered by parent liability first, then by pre-mRNA sites, then by distinct gene loci, with the gap-level margin breaking ties — and it is Table 4 the reagents of §5.1 are taken from. Each candidate was screened against all six parent transcripts rather than the two parents of its own fusion, because the FET-family donors (FUS, EWSR1 and TAF15) are paralogues with similar low-complexity amino-termini.

Specificity screening. Five screens were applied. Each is named below and referred to by that name throughout, because each reaches a compartment the others cannot and each is blind to something another catches. No single screen supports any claim here on its own.

1. **The alignment screen.** Each target window was queried against human RefSeq RNA[26] with BLAST+[27] (blastn-short, low-complexity filter off, ≥14/16 identity). A transcript window matching a design at 14 or more of its 16 positions is a *near-match*, and every near-match was classified by whether the six-nucleotide gap was itself base-paired. One that pairs the gap is *gap-paired*, or equivalently gap-spanning, and RNase-H1 could cleave there; one that pairs only the wings could not be cleaved. This is a heuristic search, and it retains only a limited number of hits per query. That hit cap makes every count taken at the default depth a lower bound rather than a total, and §3.10 measures by how much. Records of the six parent genes are counted separately and excluded from every near-match count reported here, since each parent pairs one wing by construction and would otherwise dominate the list. The parents are assessed instead by the gap-level margin and by screen 4.

2. **The exhaustive transcript scan.** A seed-and-extend scan searched 186,185 transcripts (GRCh38.p14) for exact and ≤1-mismatch matches. It is complete for substitutions by construction, reads the sense orientation only, and does not detect insertions or deletions.

3. **The pre-mRNA screen.** Screens 1 and 2 search mature transcript, so a third covers the nuclear compartment they cannot reach. Unspliced sequence and exon coordinates for all six parents were retrieved from Ensembl, and every target window was scanned against them in both orientations at ≤2 mismatches. That is the threshold the alignment screen admits, which keeps the two comparable: a stricter one here would return a cleaner pre-mRNA result for that reason alone. This arm is exhaustive for substitutions, seeded on three blocks of the 16-mer so that a hit within the threshold must match one block exactly. Each hit is classified as wholly intronic, wholly exonic, or spanning an intron–exon boundary, since only the exonic class could have been visible to a transcript screen.

4. **The mature-parent screen.** This addresses the parent transcripts screen 1 excludes, in the compartment screen 3 cannot reach. Mature parent transcripts were spliced from the same Ensembl records, and every target window was compared to every 16-nucleotide window of all six, forward orientation only. A window counts only if all six gap positions are paired. Its size is the longest contiguous run of perfect pairing containing the whole gap, which is the duplex RNase-H1 would see. Runs shorter than ten base pairs are not treated as plausible substrates. That is a stated threshold, not a measured one, so every design's longest run is released. Ten is the strict end of a figure its source gives as a possible explanation of its own observation — "this could be because RNase H1 requires a minimum length of 7 to 10 RNA:DNA hybridized nucleotides to bind with its hybrid binding domain" — rather than as a measured minimum.[28] The qualifier does not transfer cleanly: the run counted here is the whole contiguous duplex, of which exactly six nucleotides are the RNA:DNA pairs of the gap and the rest are LNA:RNA wing pairs the source's wording would not count. Ten here is therefore a total-duplex length, while the source's 7 to 10 is a count of RNA:DNA nucleotides that this architecture holds at six in every design, below

the range the source states; what relation the one criterion bears to the other is not established here. The count it produces is a floor: at seven the same screen returns 175 of 190 rather than 87.

The null for this screen. Unlike the genome scan, this one is bounded by six transcripts and so has no chance expectation of its own, and a count against six transcripts is not interpretable without one. Six ensembles of arbitrary 16-mers were therefore pushed through the identical screen, 200 draws per design in each: each design's target window shuffled to preserve base composition; shuffled to preserve dinucleotide composition by an Eulerian-path shuffle; drawn from uniform bases; drawn from the panel's pooled base composition; and two arms holding either the catalytic gap or the wings fixed while shuffling the other. A seventh is not a chance null but a structural one: a random window of a real donor parent joined to a random window of real *NR4A3* at one of the junction offsets the panel's own registers use, which keeps the whole design rule and randomises only where in each transcript the two pieces come from. Proportions carry Wilson 95% intervals. The pseudo-random stream is written out in the released code rather than taken from the interpreter's own, so the artefact is reproducible bit for bit. §3.5 reports the result.

5. **The genome scan.** Screens 1 to 4 are bounded either by an annotation or by six transcripts. The fifth removes that bound. Every distinct target window and its reverse complement was tested against every position of GRCh38 in both orientations at ≤2 mismatches, exhaustively: 2,948,609,696 windows over a measured 3.10×10^9 nucleotides, with no seed, no word size and therefore no search sensitivity to quantify. §3.6 reports it.

Strand orientation. A match matters only if an antisense oligonucleotide could base-pair with it, which means a match on the sense strand. One carrying the reverse complement of the target window is not a liability at all. `blastn` searches both strands, so a reverse-complement hit passes an identity filter unless orientation is parsed, and screens produced before that parsing was added recorded such hits as cleavage risks. Orientation is now parsed and filtered in all 38 junction screens and the 183 designs they hold, and therefore in every cleanliness statement made here. The only two screens in the released set that are not filtered are modelled control junctions built in amino-acid rather than transcript coordinates, which carry no junction and support no claim. The same rule governs pre-mRNA, which is transcribed in transcript orientation: a forward match can be base-paired and a reverse-complement match cannot.

Designs are called clean below when they carry no sense-strand near-match, and that is always a statement about a complete hit list at a stated search depth. Both qualifications matter. A hit list the cap truncated is not complete, so no verdict is available for that design, and a design clean at one depth need not be clean at another. §3.4 and §3.10 report both effects.

Target-site accessibility. Estimated as mean unpaired probability over a local fold of up to 180 nucleotides, computed with the ViennaRNA partition function.[29] It spans 0.160 to 0.707 across all 190 designs at real exon junctions, with a median of 0.477. It is released with the artefacts and is not used to rank anything here, and the omission is deliberate rather than an oversight. Three reasons, in decreasing order of force. First, the quantity is not the one the paper is about: accessibility predicts whether an oligonucleotide can reach its site on the fusion, and every question asked here is whether it can be told apart from a parent once it does. An inaccessible site is a potency problem, and potency is not claimed for any sequence. Second, the estimate is a fold of a naked transcript, whereas the compartment that matters is a nascent, protein-coated pre-mRNA; measured antisense activity correlates with such predictions weakly and inconsistently, which is why the field selects by screening rather than by folding. Third, it would have no purchase on the result even if it were reliable: the surviving candidates are separated by orders of magnitude of predicted off-target load, and reordering them on a quantity that spans 0.160 to 0.707 across the panel would substitute a weak predictor for a strong one. The values are released so that a laboratory ordering several oligonucleotides at one junction can break a tie on them, which is the use they support.

Expression of the off-target loci. No screen above says whether a matched gene is transcribed where the drug goes. For four of the five junctions with a published exon-resolved breakpoint — those Table 6 covers — the gene loci their deeper screens return in the gap-paired class were read against GTEx v8 median TPM.[30] No such reading was taken at *TFG* exon 7, so that junction carries no expression reading rather than a negative one. The readings are in two blocks, reported separately and never combined, and are described against two cuts used for legibility rather than as thresholds of concern: a lower cut of 1 TPM, below which a reading is taken as below detection, and an upper cut of 10 TPM. The first block is liver and both kidney compartments, the organs a systemically dosed phosphorothioate gapmer distributes to. The second is six soft-tissue types, standing in for the compartment EMC arises in, since no atlas contains the tumour itself. NCBI Gene supplied locus identity, so a locus with no reading is attributed rather than left blank. The Human Protein Atlas[31] was read as a transport check only, its consensus incorporating GTEx rather than confirming it independently.

Discrimination model. The binary assumption that any mismatch inside the gap abolishes cleavage is not supported by the primary literature and is not used for any claim of cleanliness. The field's general figure for single-nucleotide discrimination by a gapmer carrying no positional modification in its gap is approximately five-fold,[32] and at 16-mer length one study reports no efficient discrimination at all.[33] Both are measured against a single-nucleotide substitution rather than a fusion junction, and the pessimistic one used unmodified antisense DNA. They are used here as bounds for unmodified chemistry, not as a property of this architecture. Gapmer-specific work points the same way, and is the reason the bounds are not narrowed. Across more than 120 gapmers spanning five single-nucleotide changes, only two or three achieved preferential cleavage of the mutant allele in cells.[34] Where allele selectivity is achieved, it is engineered by modifying a gap position to block cleavage of the near-match, rather than obtained from the mismatch itself.[35]

Every screen that resolves the gap was therefore re-scored under both bounds as a graded residual cleavage load, holding the hit set fixed so that only the scoring changed: all 38 junction screens, and 39 of the 93 screens released in total. The exceptions are one coverage-only control screen that records no gap-mismatch depth and so cannot be graded at all, and the 53 deeper re-screens, which are released ungraded. The re-score counts only sense-strand hits where it can, meaning where the retained hit list is complete and every hit's strand is therefore known.

Two distinct bounds follow, and they run in opposite directions. Where a hit list is truncated, the strand of the remainder is unrecoverable, so some designs keep a strand-blind count. That over-counts liability, because it includes matches no antisense oligonucleotide can hybridise, and is an upper bound. Screens produced before the orientation fix carry that count for every truncated design; later screens carry an already-filtered one. The same truncation also means fewer hits are recorded than the search returned, so the count of hits is itself right-censored: a lower bound on how many exist. Each design records which bounds apply to it.

Duplex thermodynamics. A base count is a proxy for discrimination, and a free energy is the field's standard instrument for it, so each design was also scored thermodynamically. A junction gapmer is a perfect complement of the fusion across all 16 positions, while a parent transcript can pair only the half of the oligonucleotide it contributes. The comparison is therefore the full 16-mer duplex against the donor-side and acceptor-side runs alone. Nearest-neighbour enthalpies and entropies for a DNA:RNA hybrid were taken from Sugimoto and colleagues,[36] and $\Delta G°37$ computed as $\Delta H° - T\Delta S°$. The 250 nM strand concentration enters only the melting temperature used to check the arithmetic against an independent implementation, which agreed exactly. That check verifies the summation, not the choice of strand, which is fixed by the table's documented convention that the sequence supplied is the RNA one.

These designs carry LNA wings and the table is for an unmodified hybrid, so what is computed is the duplex the DNA backbone would form. Because the junction lies inside the gap, each parent pairs exactly one of the two five-nucleotide LNA wings while the fusion pairs both. LNA should therefore widen this margin rather than narrow it, which makes the value reported a conservative floor. That direction follows from the architecture and was not computed: no LNA parameters were applied.

Conventional design rules. Every design was separately audited against four conventional antisense design rules: GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, and no CpG dinucleotide. The audit is not there to grade the designs, but to ask whether conventional triage and the gap-level margin would select the same molecules.

Availability. All code, graded artefacts and per-design tables are released under a single archived version, deposited from the public repository at `github.com/trimcrae/Rare-cancers` [ARCHIVE DOI]. Every result reported here is re-derived from the committed artefacts in that archive without network access or credentials. Regenerating the specificity screens from scratch is not offline, because the alignment screen queries NCBI BLAST and the exhaustive transcript scan downloads the GRCh38.p14 RefSeq RNA set. No reported number requires it, because each screen's hit set is archived and the re-scores hold that hit set fixed. The pre-mRNA and mature-parent screens are fully offline against the archive: the retrieved unspliced sequence and exon coordinates travel with it.

3 · Results

The results are ordered for a laboratory deciding what to make. Sections 3.1 to 3.3 ask whether designs exist at all, whether one can serve more than one patient group, and whether the partner gene predicts a clean design. Section 3.4 gives the designs that survive the transcriptome screens, 3.5 what they would do to the parent genes and how that compares with what arbitrary sequence does, and 3.6 the candidates left once every screen has been applied. Sections 3.7 to 3.9 carry what a bench decision turns on next: where the off-target loci are expressed, which catalytic gap to build, and whether duplex free energy or the conventional design rules would select different molecules. Section 3.10 states how far the counts themselves can be trusted, and it bounds everything above it.

How the counts are denominated. Six numbers recur below and they are not interchangeable. 231 is the donor-exon by acceptor-exon pairs graded for frame. 38 are the in-frame junctions among them. Those 38 carry 190 design records, which are 176 distinct molecules, because nine of the 16-mers span more than one partner's junction and are recorded once per junction. Of the 190, 183 have a returned specificity screen; the other seven failed at the remote service, which matters because a free energy needs only a sequence where a screen needs a query that came back. 187 is the count re-screened at the tenfold deeper ceiling. Each result below names the denominator it uses.

3.1 · The reading frame as the bound on junction space

Grading all 231 donor-exon by acceptor-exon pairs across the five partners returns 38 in-frame junctions (Table 1, Figure 1). The refusals are structural. *NR4A3* exon 2 carries no coding sequence and is refused in every pair, and an exon-4 acceptor would delete the *NR4A3* DNA-binding domain that every reported EMC chimera retains. All the variance therefore sits in the exon-3 column. Within that column, being in frame reduces to a single arithmetic condition, a donor coding phase of 1, which is necessary and sufficient across its 77 rows but only necessary across all 231.

Among these, *EWSR1* exon 12 joined to *NR4A3* exon 3 is the junction reported most often: type 1 in 10 of the 15 *EWSR1*-rearranged tumours of an 18-case series.[37] Designs at this junction therefore correspond to the largest documented patient group.

No design at any of the 38 junctions is a perfect complement of any of the six parent transcripts. That test excluded none of the 190, because a junction-spanning window cannot occur intact in a parent. GC runs 25.0–75.0% across partners, with 132 of the 190 inside the conventional 40–60% band. Finding candidate sequences is therefore not the constraint on this modality in this disease.

3.2 · Cross-partner coverage by a single oligonucleotide

Nine designs span the junction of more than one junction exactly, and all nine draw from *EWSR1*, *TAF15* and *FUS* (Figure 3). Five cover the same three-partner set, differing only in register across the junction. The best by gap-level margin is 5′-GGGCATATCATCAAAC-3′ (43.8% GC, gap-level margin 3), which divides eight donor and eight acceptor bases at the junction of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, and occurs in none of the six wild-type parent transcripts. The basis is sequence identity: the three donors are identical over the ten bases immediately 5′ of their breakpoints, diverging at the eleventh. No design draws more than ten donor bases, which is what makes the coverage arithmetically possible.

In one respect the published data contradict the clinical reading of this result. The only exon-resolved *TAF15::NR4A3* breakpoints reported in EMC are at exon 6, not exon 11. The primary report of the variant fusion places the breakpoint there,[38] and in a series of 18 EMCs all three *TAF15*-rearranged tumours carried exon 6 joined to *NR4A3* exon 3.[37] The exon-6 junction shares a single donor base with the exon-11 junction, so this oligonucleotide cannot engage the *TAF15* junction that patients are reported to carry.

That junction is itself in-frame and yields five fusion-specific designs (43.8–50.0% GC), all five screened and orientation-filtered. Every one of them retains a sense-strand near-match spanning the catalytic gap. At the tenfold deeper ceiling, where every hit list is complete, those recount to three gene loci at best, and five for the design its gap-level margin ranks first, three of those five annotated only as predicted gene models (Table 4). Two of the five nonetheless return no exact and no single-mismatch match on the exhaustive transcript scan.

So the one *TAF15* junction with a published breakpoint is designable and is not among the cleaner junctions, while the junction the multi-partner result rests on has no reported patient. For *FUS* no exon-resolved EMC breakpoint has been published at all. The three-partner result is therefore a statement about FET-family sequence architecture and a hypothesis about junctions not yet observed. It is not a claim that one reagent serves three patient groups. Testing it requires breakpoint sequencing of archival *TAF15*- and *FUS*-positive cases.

3.3 · The non-FET partners: coverage and specificity

TCF12 and *TFG* are the partners in this panel that are not FET-family proteins, and neither appears in any of the nine multi-partner sets: all nine draw only on *EWSR1*, *TAF15* and *FUS*. *TCF12* reaches multi-partner coverage only under a relaxed criterion that tolerates mismatches in the oligonucleotide wings. That check had little power to fail, since any non-homologous donor would be excluded, so it does not separate FET paralogy from incidental exon homology. The stronger evidence for paralogy is that four additional two-partner sets are also FET-only.

Specificity does not sort by partner (Table 2). Taking at each junction the lowest count any of its designs achieves after the orientation filter, every one of the five partners has at least one junction whose best design carries no sense-strand near-match across the catalytic gap **at the default search ceiling**: three of eight at both *TCF12* and *FUS*, two of eight at *EWSR1*, one of eight at *TAF15* and one of six at *TFG*. At the tenfold deeper ceiling that becomes four partners: each of *EWSR1*, *FUS*, *TAF15* and *TCF12* keeps one such junction and *TFG* keeps none, its single default-depth zero returning 29 across the gap when searched deeper. The minima therefore separate junctions rather than partners. Which exon a fusion breaks at matters more for specificity than which gene it breaks into.

The same tension the *TAF15* result carries applies to *TCF12*, and in the same direction. The one published *TCF12::NR4A3* breakpoint describes a chimera retaining the first 108 TCF12 residues,[5] and names no exon; the same authors deposited the chimeric cDNA, and that deposit resolves the junction to the nucleotide. GenBank AF289510.1 is 421 bases long and carries two chromosome-tagged source features that meet at the junction, bases 1 to 263 on chromosome 15 and bases 264 to 421 on chromosome 9. Mapped against the transcript models used here, the donor side ends at *TCF12* exon 5 and at no other exon, the acceptor side begins at *NR4A3* exon 3 and at no other exon, and the twelve bases either side of the seam are identical to the seam the panel was designed on. Translating the deposit reproduces its own recorded protein, which is a substring of the modelled chimera. That junction is in-frame and designable, and its best-margin design retains 17 gap-spanning near-matches at the deeper ceiling, every one of them a variant of a single curated locus, *PIK3CG* (Table 4). None of the four *TCF12* designs with no sense-strand near-match is at that exon. So for *TCF12* as for *TAF15*, the junction a patient is reported to carry is designable and is not among the clean ones, while the clean junctions have no reported patient. What remains unmeasured at *TCF12* is not the exon but the distribution: one *TCF12*-rearranged tumour has been sequenced at this junction, and it is the tumour the junction was defined by. That gap is a property of how the disease is diagnosed rather than of the retrieval: *TCF12*-rearranged tumours have been counted in several independent cohorts and sequenced in one of them, because the break-apart assay those cohorts used to identify them reports that the *NR4A3* locus is split[6] without locating the seam within it. Whether this exon recurs is therefore untested rather than refuted, and no search of the nucleotide or read archives returns a second *TCF12::NR4A3* sequence.

The same archive search resolves *TFG*, the other non-FET partner, in the same way and to the same limit. No paper places a *TFG::NR4A3* breakpoint at an exon, but a deposited chimeric mRNA record does — GenBank AY532911.1, annotated as a *TFG-NR4A3* fusion protein — and it lands on *TFG* exon 7 joined to *NR4A3* exon 3, a junction this panel already carries. Four patent sequence records agree with it base for base at the seam, and they corroborate a sequence rather than four patients, being one family from one group. As at *TCF12*, what the deposit supplies is the exon and not the distribution: no source states what fraction of *TFG*-rearranged tumours break there, and *TFG* does not appear in the partner counts of the 58-case cohort every coverage figure here is denominated on, so this changes which junctions are reported and no percentage.

3.4 · Strand orientation, and designs with no sense-strand near-match

All 38 in-frame junctions were screened with orientation filtered, covering 183 designs, and Table 2 gives the per-junction result. Of the 1,677 apparent cleavage risks across the retained hit lists, 738 sit on the minus strand, or 44%. An antisense oligonucleotide cannot base-pair with those at all.

The proportion is not uniform. It runs from 0% at *TFG* exon 4, where no apparent risk is minus-strand, to 100% at both *EWSR1* exon 1 and *TCF12* exon 7, where every one is. That non-uniformity is what makes the filter worth applying rather than approximating. A uniform inflation would rescale every junction and leave their ordering intact; this one reorders them. *EWSR1* exons 7 and 13 return 55 and 57 apparent gap-spanning hits, and after filtering they stand at 6 and 53.

After filtering, nine designs at six junctions carry no sense-strand near-match among non-parent transcripts (Table 3), spanning four of the five partners: three at *EWSR1* exon 1 (5′-GGGCATATCCGTGGAC-3′, 5′-GGCATATCCGTGGACG-3′, 5′-GCATATCCGTGGACGC-3′), one at *FUS* exon 8 (5′-AGGGCATATCGGAGTC-3′), one at *TAF15* exon 1 (5′-GGGCATATCCGACATG-3′), and four at *TCF12* — 5′-GGGCATATCTCTATAA-3′ at exon 17, 5′-CAGGGCATATCTTGCA-3′ at exon 9, and 5′-GGCATATCAAGCGCTG-3′ and 5′-GCATATCAAGCGCTGC-3′ at exon 7. The exhaustive transcript scan agrees independently: each returns no exact and no single-mismatch match anywhere in 186,185 transcripts. The two screens fail in different ways, so their agreement is not a restatement. One is a heuristic alignment search over both strands; the other an exhaustive substitution scan over the sense orientation only. The pre-mRNA screen, over a compartment neither of those reaches, does not overturn them either: none of the nine has a sense-strand site in parent pre-mRNA (§3.5).

The graded re-score agrees, with one instructive exception. Scoring every retained hit by the residual cleavage a gap-internal mismatch is predicted to permit, under both literature bounds, returns a residual load of zero for all nine. It returns zero for one further design too, at *FUS* exon 11, which is not counted as clean here. That design returns 21 near-matches, of which only 15 are retained, and all 15 are minus-strand. The graded score therefore sees nothing it can score, while the cleanliness criterion refuses the design because the strand of the six unretained hits is unknown. The graded model has no censoring guard, so it can award a zero the hit list does not support, and the stricter count is the one reported. A zero for the nine is arithmetic rather than an independent measurement: it follows from their having no sense-strand hit to score.

Every junction was then re-screened at a tenfold deeper alignment ceiling, with retention raised to match so that no hit list is truncated: 38 junctions and 187 design records. The result withdraws most of the set above. The 187 are the panel's 190 less three that failed at the remote service on this pass, two at *FUS* exon 5 and one at *TFG* exon 2. Each had already returned 23, 41 and 31 near-matches at the default depth, so none was a candidate and no count below depends on them. Only three of the nine still carry no sense-strand near-match: 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8, 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 and 5′-GGCATATCAAGCGCTG-3′ at *TCF12* exon 7, each of which returned the same count at both depths. The other six did not. The three *EWSR1* exon-1 designs had returned no near-match at all at the default ceiling and return 27, 29 and 84; 5′-GGGCATATCTCTATAA-3′ at *TCF12* exon 17 goes from 8 to 118, and 5′-CAGGGCATATCTTGCA-3′ at *TCF12* exon 9 from 7 to 67. Three of the six carry hits that span the catalytic gap and so are cleavage risks rather than merely sense-strand matches: 64 for 5′-GCATATCCGTGGACGC-3′, 14 for 5′-GGGCATATCTCTATAA-3′ and 11 for 5′-CAGGGCATATCTTGCA-3′. A count of zero at the default ceiling was not a count of zero, which is the sharpest form of the bound §3.10 sets out.

The deeper pass also decided what the default one could not. Seven of the 190 designs had failed at the remote service and carried no count at all; all seven returned at the deeper ceiling, six of them dirty and one — 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7 — with three near-matches and none on the sense strand. So the set of designs with a complete hit list and no sense-strand near-match is four at this depth rather than three: a design the shallower pass never screened joins the three that survived it. The deeper counts are reported as their own measurement and no figure quoted above is restated from them.

The orientation call is corroborated independently of any of this. Ten designs return perfect 16/16 BLAST matches while the sense-only exhaustive scan reports no exact match. Both results can only be correct if every one of those BLAST hits is on the minus strand, and every one is.

3.5 · The parents: liability in pre-mRNA and in mature transcript

RNase-H1 is active in the nucleus and gapmers engage pre-mRNA, so a screen over mature transcripts cannot see intronic or intron–exon-spanning sites. That omission is not neutral in its direction. A junction gapmer's two halves are both exonic, and in a parent pre-mRNA an exon is followed by an intron rather than by the next exon. Parent pre-mRNA is therefore precisely where a design's donor half sits beside sequence no mature screen has compared it against. A mature-only screen returns a low count partly by construction.

Of the 190 designs, 53 have a near-match somewhere in parent pre-mRNA. Nineteen carry one that meets all three conditions that would make it dangerous: it is on the sense strand, it pairs the catalytic gap in full, and it touches intronic sequence. That third condition is what makes such a site invisible to both transcript screens, rather than a re-count of something already reported.

The step from 53 to 19 is a threshold rather than a measurement, and the class it removes is the one the Methods decline to dismiss. Forty designs carry a sense-strand parent pre-mRNA site. The 19 counted here are those pairing the catalytic gap in full; the remaining 21 pair all of it but one or two positions. Of their 28 sites, 26 are a single gap mismatch short, and five are in *NR4A3* itself. Under the bounds this work adopts, a single mismatch inside the gap does not abolish cleavage, so those 21 are not a null result. They are excluded because a graded count over this compartment would need a discrimination model the literature does not supply for a parent duplex. The same condition governs the mature-parent screen below, which considers only windows pairing the whole gap. Every count in this section should be read as the fully-paired class, not as the whole parent liability.

Those 19 sites fall into two classes that do not mix, and only one is mechanistically interesting. Nine are intron–exon-spanning, and every one is in *NR4A3* at the same place: six or seven nucleotides into intron 2, spanning the boundary into exon 3. That follows from the design problem. A junction gapmer's acceptor half is the 5′ end of *NR4A3* exon 3, and the wild-type *NR4A3* transcript reaches that same exon across its own splice junction. So a design whose donor half also matches the 3′ end of intron 2, within the mismatch budget, pairs across the real splice site. That is a route to wild-type *NR4A3* engagement which does not pass through the fusion at all, in the compartment where RNase-H1 is active. It is the discrimination question this paper is about. The other ten are wholly intronic and every one is in *TCF12*, which contributes 365,096 of the 517,157 intronic nucleotides searched. That is 71% of the search space accounting for 100% of the class, which is what sequence volume alone predicts and should not be read as anything about *TCF12*.

The liability tracks the tiling register, of which the gap-level margin is a function. At margin 1, 12 of 76 designs carry a pre-mRNA site; at margin 2, 7 of 76; at margin 3, none of 38. Eight of the nine *NR4A3* boundary sites are at the shortest donor-side register, which needs the fewest intronic bases to match. None of the nine designs with no sense-strand near-match on either transcript screen carries one.

The second class is in mature parent transcript, and it is larger. Each of the first three screens misses it for its own reason. The alignment screen excludes parent records by design and filters at ≥14/16 identity. The exhaustive transcript scan admits only one mismatch. The pre-mRNA screen searches unspliced sequence and so cannot reach a mature exon–exon junction. A parent duplex of 11 or 12 contiguous base pairs that pairs the whole catalytic gap is therefore invisible to all three, while being exactly what RNase-H1 requires.

Screen 4 compares every design's target window to every window of all six mature parents. Of the 190 designs, 87 have a duplex of at least ten base pairs, and 61 of those 87 are against wild-type *NR4A3*, the transcript this modality must spare. Those 61 are not 61 distinct sites: 59 of them are the same one, the mature exon-2/exon-3 seam every design's acceptor half reaches, which is the mature-transcript counterpart of the pre-mRNA concentration above. A 62nd pairs *NR4A3* at eleven base pairs but another parent at twelve, so it is attributed elsewhere. The count falls steeply with the gap-level margin: 50 of 76 designs at margin 1, 29 of 76 at margin 2, and 8 of 38 at margin 3. That is what the margin's definition predicts, since at margin 1 a parent needs one lucky base to pair the whole gap and at margin 3 it needs three. Five of the nine designs of §3.4 carry such a duplex at 11 or 12 base pairs, including 5′-CAGGGCATATCTTGCA-3′ against wild-type *NR4A3* itself. The margin is therefore a predictor of parent engagement rather than a guarantee against it, because it counts bases unique to the fusion at the junction without asking whether a parent carries them elsewhere.

Eighty-seven of 190 is 45.8%, with a nominal binomial Wilson 95% interval of 38.9–52.9% — nominal because it treats the 190 records as independent draws, which the close of this section states they are not, so it is narrower than a junction-clustered interval on the same counts. A count of that kind means little without a null, so the same screen was run over arbitrary 16-mers. Only the query changes: the same six mature parents, the same forward orientation, the same ten-base-pair threshold. Scrambling each design's own target window, which preserves its base composition and is the scrambled-gapmer control §5.4 asks a laboratory to make, gives 6.2% (5.9–6.4%); a dinucleotide-preserving shuffle gives 10.0% (9.7–10.3%); 16-mers drawn from uniform bases give 6.9% (6.7–7.2%), and from the panel's pooled base composition 7.2% (6.9–7.4%). A calculation agrees with the sampled figure rather than the sampled figure standing alone: the gap must pair, at 4^{-6}, and the run must then extend four further nucleotides across the two wings, at 1/64, which over the 19,921 parent windows searched predicts 7.3%. The observed rate is about sevenfold that, and the arm the modality actually turns on separates further still: 32.1% of designs pair the gap against wild-type *NR4A3* specifically, against 1.8% of scrambles. Scrambles are the weakest null run here; against the structural chimera null below, the strictest, that same arm separates 32.1% from 9.3%.

A second null asks whether that excess is a fact about reported breakpoints or merely about the design rule. Joining a random window of a real donor parent to a random window of real *NR4A3*, split at a junction offset the panel's registers use, reproduces everything the rule specifies, namely donor sequence 5′, *NR4A3* sequence 3′ and the junction inside the catalytic gap, while destroying only the fact that the two pieces meet where a tumour joins them. Those chimeras reach 23.8% (23.3–24.2%). At the ten-base-pair threshold, therefore, about half the observed liability is generic to any chimera of these two transcripts, and about half is specific to the real junctions. That apportionment is a property of the threshold rather than a stable share: across the seven-to-ten window the threshold comes from, the generic half runs from 52% at ten to 94% at seven, so at the loose end of the cited range almost all of the liability is what any parent-to-parent chimera gives. Two further arms locate it no more finely: holding the six gap bases and scrambling the wings gives 9.1%, and the mirror gives 8.8%, because a run reaching ten base pairs needs the real gap and the real flanks together. None of these rates is a significance test and none is offered as one. The 190 records are 176 distinct molecules tiled at overlapping registers across 38 junctions, so they are not independent draws, and a test treating them as 190 would be wrong about its own denominator.

3.6 · The surviving candidates, and a genome-wide check

Composing this with the deeper re-screen of §3.4 leaves three candidates in the whole panel, and they are not equally secure. 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8 and 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 carry no sense-strand near-match at ten times the default search depth, no single-mismatch match on the exhaustive transcript scan, no pre-mRNA site and no mature-parent duplex. Neither depends on the ten-base-pair threshold, because no window of any parent pairs their catalytic gap at any length: their longest run is zero rather than merely short. 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7 passes the same screens, but not in the same way. Its longest parent run is eight base pairs against wild-type *NR4A3*, which is below the threshold rather than absent, so it is a candidate at the stated cut and not at a stricter one. The fourth design with a clean deep screen, 5′-GGCATATCAAGCGCTG-3′ at the same junction, is excluded by an eleven-base-pair duplex with wild-type *TCF12* — its own donor parent, not the acceptor. That is the honest size of the candidate set, and no junction among them has a published patient breakpoint — the exon-resolved *TAF15* breakpoints reported in EMC are exon 6, and for *FUS* and *TCF12* exon 7 none has been published at all. Selecting within each junction rather than across the panel changes what is available, not what is clean: Table 4 applies the same criteria at all 38, where 35 have a design that clears the parent screen and *TAF15* exon 14, *TCF12* exon 3 and *TFG* exon 2 have none. All five junctions of the panel with a published exon-resolved breakpoint have one: four of them at the top gap-level margin, with longest parent runs of eight, eight, nine and seven base pairs, and *TFG* exon 7 at a margin of two with a longest parent run of nine.

Both classes were bounded the same way: exhaustive over six parent transcripts and silent about every other gene. The genome scan, screen 5, removes that bound.

A raw genome-wide count is not a result at this threshold. Chance alone predicts of order 10^3 near-matches per 16-mer over a genome for any 16-mer whatever, so the informative readings are stratified. Exact 16/16 matches are the class where chance expectation is of order one: 1.37 expected per design against 236 observed across 176 windows, which is at chance. Load relative to that expectation separates designs where a total cannot — the median design sits at 0.98 of its expectation and 14 of 176 exceed twice it. And the repeat split, free from a soft-masked reference, shows 52.5% of hits fully repeat-masked against a genome that is 51.4% masked, so the load is not repeat-driven.

The decisive reading is a lookup rather than a count: does any design have a gap-paired, strand-agreeing site in *NR4A3*, in a parent gene, or in an *NR4A* paralogue anywhere in the genome? Twenty of 176 do. No candidate above is among them, and the two secure at any parent-duplex threshold carry a load well below chance — 0.33 and 0.24 of expectation at ≤2 mismatches, and 0.06 and 0.04 for gap-paired sites, ranking 26th and 13th of 176. That is the strongest statement this work can make about them, and it is a statement about predicted hybridisation and not about cleavage.

3.7 · Expression of the off-target loci

No screen above establishes that a design's off-target gene is transcribed where the drug goes, and that discount applies to every count in this paper. Read against reference expression data, the gap-paired loci of the best design at each of the four junctions Table 6 covers separate, in the direction opposite to the sizes of their loads. The *EWSR1* exon 12 reagent's six loci carry 123 of the panel's 649 transcript records, and none of the four measurable ones reaches the upper cut in liver or either kidney compartment. *ANKS1B* supplies 67 of them and sits below the lower cut in all three, peaking instead in brain at 24.9 TPM.

The *EWSR1* exon 13 reagent inverts that comparison, and the inversion is the reading a person choosing between them needs. Its load is the smaller of the two by locus count, two against six, but both of its loci are transcribed in the compartments a systemic phosphorothioate gapmer distributes to: *CDC42SE1* reaches 78.97 TPM in kidney medulla and 22.7 in liver, and *FNBP1* reaches 14.06 in kidney medulla on 42 transcript records. The exon 12 reagent's larger load sits away from those compartments, its brain- and gut-restricted loci reaching no measurable value at the upper cut in liver or either kidney block. So the two axes disagree, and for a reader deciding which oligonucleotide to make, the exposure axis is the one that bears on where an off-target effect would be delivered, while the count is an annotation property of the loci returned. Neither axis is a risk ranking: every hit behind both is a 14/16 match, no cleavage is predicted anywhere here, and an expressed gene is necessary but not sufficient for an effect. What the comparison does not touch is the reason to make the exon 13 reagent at all, which is patient coverage: it is the second-most-common *EWSR1* transcript and takes the reagent set from 68.4% to 79.0%. The *TAF15* exon 6 reagent returns five loci. Of these, *NRP1* reaches 6.6 to 17.8 TPM across all three exposure tissues and is the only one all five of that junction's tiling registers return, on five transcript records, so robustness to register and record count order it differently. The tumour-compartment proxy orders them differently again, *HNRNPA2B1* carrying the panel's highest value there at 656.6 TPM in tibial nerve, ahead of *LAMA4* at 268.6 TPM in cultured fibroblasts.

3.8 · Gap length trades junction specificity against parent-duplex competence

The panel above is one geometry. Tiling the same junctions at 5-8-5 and 5-10-5, wing fixed at five nucleotides, resolves what a longer catalytic gap buys and what it costs (Table 5, Figure 2).

What a longer gap buys and what it costs are the same nucleotide. Inside the gap, the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap. That holds for every design in all three panels rather than on average, and it fixes the direction of both trades. Within one geometry the gap is fixed, so the two move inversely: at 5-6-5 the 38 designs at margin 3 concede three nucleotides of contiguous parent-paired gap DNA, the 76 at margin 2 concede four and the 76 at margin 1 concede five. What the identity forbids is raising the margin past a geometry's ceiling, which is half its gap rounded down; that takes a longer gap, and a longer gap raises the parent-paired run at every register, the best-margin design running from margin 3 conceding 3 nucleotides at 5-6-5 to margin 5 conceding 5 at 5-10-5. Lengthening the gap therefore makes RNase-H1 more competent against the fusion and against the parent together, and no register choice within a geometry escapes that, though it does trade the two against each other.

Both directions are large. The best available gap-level margin rises from 3 to 4 to 5, and the junction-spanning registers per junction from five to seven to nine. At the *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 junction, the design carrying that margin sheds its transcriptome load completely: 123 sense-strand cleavage risks across the gap at six gene loci become 3 at one locus and then none. Over the six junctions screened at every geometry, designs carrying no such risk rise from 8 of 30 to 28 of 42 to 54 of 54, and the most risk loci on any one design falls from seven to two to none.

Against that, the contiguous DNA a wild-type parent pairs at the same junction rises from 3 to 4 to 5 nucleotides, and the most stable parent duplex from −7.77 to −8.66 to −10.25 kcal/mol. The corpus shows the same trade. Designs whose parent pairs at least five nucleotides of contiguous gap DNA, the shorter of the two reported minima for RNase-H1, rise from 76 of 190 to 228 of 266 to 342 of 342, and the median most stable parent duplex falls from −8.66 to −14.58 kcal/mol. At 5-10-5 that count is every design, and necessarily so, since the larger half of a gap of ten cannot be under five. At 5-6-5, 114 of 190 designs keep the parent below it.

Part of the fall in near-matches is guaranteed by the instrument rather than measured. At a fixed budget of two mismatches, every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the reachable set can only shrink as the design lengthens. Two mismatches is also a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall, and which designs reach zero, are measurements. The parent-side quantities carry no such qualification, being computed from the junction rather than from a search.

Two liabilities the transcript screens do not reach appear to move the favourable way, and neither reading survives one fixed criterion. A mature parent can pair the whole gap for 181 of 190 designs at 5-6-5, 130 of 266 at 5-8-5 and 87 of 342 at 5-10-5, but the whole gap is a six-nucleotide coincidence in the first and a ten-nucleotide one in the last, so the three counts are not the same test. Held to the ten-base-pair criterion applied everywhere else here, the liability is flat: 87 of 190, 88 of 266 and 87 of 342, the two criteria coinciding at a gap of ten because the gap alone is then already a ten-base-pair hybrid. Designs pairing the gap in parent pre-mRNA fall from 19 of 190 to 9 of 342, but that arm is a search at a fixed two-mismatch budget and inherits the nesting bound above rather than the parent-side quantities' freedom from it. Nor is the effect confined to the longest geometry: 5′-CAGGGCATATCAAGCGCT-3′ at *TCF12* exon 7 returns no near-match at all, where the 16-mer surviving at that junction returns three.

3.9 · Duplex thermodynamics and conventional design rules

Scored as free energies, every one of the 190 designs favours the fusion duplex over the best duplex either parent can form, by 4.8 to 13.1 kcal/mol with a median of 9.6. The denominator is 190 here rather than the 183 of the specificity screens for the reason given at the head of §3. Every design favours the fusion because a parent pairs roughly half the oligonucleotide, and half a duplex is much the weaker one. That separates two things a base count conflates. Discrimination at the level of *binding* is not marginal here, and is not what constrains the modality. What remains unresolved is discrimination at the level of *catalysis*, where RNase-H1 requires a paired DNA gap and where the literature bounds span one- to five-fold. The thermodynamic result therefore narrows the paper's central uncertainty rather than relieving it.

The two rankings agree in direction. Grouping designs by the gap-level margin the Methods define, mean ΔΔG°37 rises monotonically with it, from 8.3 kcal/mol at margin 1 to 9.9 at margin 2 and 10.7 at margin 3. That agreement is arithmetic rather than corroboration: the longest run either parent can pair is exactly 11 minus the gap-level margin for all 190 designs, so the free energy is ordering the same quantity in kilocalories. What it adds is the size of the difference, not an independent ranking, and the same caution applies to the margin's agreement with the parent screens of §3.5.

Conventional design rules select differently, and against the paper's own candidates. Of the 190 designs, 106 satisfy all four rules; the rules bind at different rates, with every design free of a G-quadruplex motif but 13 carrying a homopolymer run of four, 43 a CpG dinucleotide and 58 falling outside the 40–60% GC window. The failures overlap, so they do not sum to the 84 designs that fail at least one. The disagreement is sharpest exactly where it matters most. Of the nine designs with no sense-strand near-match (Table 3), exactly one satisfies all four rules.

Seven contain a CpG dinucleotide, the canonical TLR9 immunostimulatory motif; four fall outside the 40–60% GC window — the three *EWSR1* exon-1 designs above it at 62.5% and 5′-GGGCATATCTCTATAA-3′ below it at 37.5%. Only 5′-CAGGGCATATCTTGCA-3′ at *TCF12* exon 9 passes every rule, and the multi-partner candidate 5′-GGGCATATCATCAAAC-3′, which is not among the nine, also passes all four. So the two filters disagree where it matters: the cleanest designs this work found are, with one exception, molecules conventional triage would flag, in six of the seven cases for a CpG a base substitution removes. Both are reported rather than composed into a single score.

3.10 · The bounds on every count: chance, censoring and gene loci

Everything above is counted with instruments that have limits, and this section states them. They do not change the ranking by gap-level margin, but they do bound what any count here means, most sharply for the clean designs of §3.4, six of which lose the property at ten times the search depth.

A chance argument bounds how surprising a low count should be, and it points the other way from the reading these numbers first invite. There are 1,129 16-mers within two substitutions of any given 16-mer, so an arbitrary transcriptome position matches at ≥14/16 with probability 2.6×10^{-7}. The span that probability multiplies is measured rather than assumed. The exhaustive transcript scan counts 718,571,139 nucleotides across its 186,185 transcripts, so the prediction is 189 near-matches for any 16-mer whatever on one strand, a single figure rather than the range an assumed span would give. The alignment screen cannot test this: its hit list is capped at 50, far below 189 on one strand and 378 on the two the search covers, so any design must return fewer whatever the transcriptome contains. The exhaustive transcript scan can test it, being complete for substitutions by construction, and on the mean it comes in at chance. At the ≤1-mismatch threshold the same span predicts 8.2 per design, against an observed mean of 9.2 over the 176 distinct oligonucleotides, a ratio of 1.12, while the median is 3. That agreement is confined to the mean, and the shape disagrees in both tails: 40 of the 176 return no match at all, where a Poisson null of the same mean predicts fewer than one across the set, and the right tail reaches 100. Real transcript sequence therefore produces a distribution an independent-uniform-base model cannot, rather than a uniform shift away from the expectation, and that null separates "more than chance" from "at chance" and nothing finer, so it does not license reading the zeros as cleanliness.

Two bounds remain on these counts and neither is corrected for. The first is the hit cap. The alignment screen returns at most 50 hits per query, and 35 of the 183 filtered designs reach that cap. A further 101 exceed the 15 hits retained per design, so 136 in all carry right-censored counts. The nine clean designs return zero to eight raw hits each, and that is not a measurement either. Re-screening at a tenfold deeper ceiling raised the count of 164 of the 180 designs screened at both depths, and 129 of those had not approached the 50-hit cap. One reporting 9 near-matches returned 34; one reporting 10 returned 110; and one reporting 15, a count the pipeline treats as a complete list, returned 204. Every alignment count taken at the default ceiling is therefore a lower bound, whether or not it reached the cap. Reaching the cap is not what censors a count, and a count of zero is not exempt, which §3.4 shows for three designs at the junction the cleanliness claim turned on.

Retention is the second bound, and the restriction it imposes was tested rather than assumed. Seven design-and-junction records have no sense-strand hit among those retained and a raw count above the retention depth, so retention alone withholds a verdict on them. Their five junctions were re-screened at the tenfold deeper ceiling, with retention raised to match. That decided six of the seven, and none of the six is clean. The counts they had been reporting were severely censored: 21 raw near-matches became 161 with 5 on the sense strand, 23 became 68 with 48, 27 became 65 with 5, 35 became 78 with 10, and 47 became 196 with 119, the last at both junctions that sequence spans. The seventh re-screen did not return, and that record remains undecided. Relaxing the censoring restriction would therefore have promoted at least six records that a deeper look shows carry sense-strand near-matches, one of them 119 of them. The nine are unchanged by the test.

These are counts of transcript records, not of genes, and the distinction cuts in the candidate's favour. RefSeq carries one accession per annotated variant, so a match to a constitutive exon of a multi-variant gene is counted once per variant. Recounting every screened hit list per gene locus, over the 44 designs of the 38 junction screens whose lists are neither truncated nor missing a locus recount, gives a median inflation of 2.25 transcript records per locus and a maximum of 11.0. A typical near-match count therefore overstates the number of distinct genes involved by rather more than twofold. The distinction also separates observed from predicted sequence: RefSeq `NM_`/`NR_` records are curated, whereas `XM_`/`XR_` are computationally predicted gene models. A design's *load* is its total predicted off-target burden, counted as near-matches. A load sitting entirely in the predicted namespace is a different kind of liability from one that matches curated transcripts. For the multi-partner candidate 5′-GGGCATATCATCAAAC-3′ both effects apply and compound with the orientation filter. Of its nine near-matches, six are on the sense strand and five of those span the catalytic gap. All five are variants of a single uncharacterised locus, LOC105374140, annotated only as predicted `XR_` models. That is one gap-spanning locus from a raw count of nine, but it is not clean of curated sequence. The sixth sense-strand near-match is *H2AP* (NM_012274), whose single mismatch falls inside the catalytic gap, so the pessimistic bound counts it in full. The same design returns no exact match and a single ≤1-mismatch match on the exhaustive transcript scan.

Re-screened at the tenfold deeper ceiling, that design returns 189 near-matches. Of these, 141 are on the sense strand and span the gap, and 123 pair the catalytic gap perfectly. Both effects above apply at that depth and neither weakens. All 123 sit at 14 of 16 identity, the loosest match the screen admits. They recount to six gene loci, of which *ANKS1B* and *ZNF667* supply 104 between them. Of the 123, 82 are `XM_`/`XR_` predicted models, and no parent transcript is among them. Depth therefore raises the raw count roughly twentyfold, and the distinct gap-spanning locus count from one to six.

4 · Discussion

Designability is not the constraint. Junction-spanning designs exist at every in-frame NR4A3 fusion junction, though at three of them every design pairs a wild-type parent through the catalytic gap. Nor does specificity sort by partner. With all 38 junctions screened, four of the five partners have a junction whose best design carries no sense-strand near-match across the gap at the deeper ceiling, and all five do at the default one. It is therefore the exon a fusion breaks at, not the gene it breaks into, that predicts a clean design — and the count of clean junctions is itself a function of search depth, as §3.10 sets out.

Clean designs are much scarcer than the default search depth implies, and two independent findings converge on that. One is search depth, measured in §3.4 and bounded corpus-wide in §3.10. The other is invisible to depth at any setting: five of the nine designs form an eleven- or twelve-base-pair duplex with a mature wild-type parent that pairs the whole catalytic gap, one of them with *NR4A3* itself, where no screen filtering on global identity can see it. Three designs survive every screen applied here, two of them at any parent-duplex threshold. That is the honest size of the candidate set.

The limiting step is discrimination between the fusion and its parents, and it is not resolved here. Both cited bounds are measured against a single substitution in an otherwise fully paired duplex. Neither transfers to a parent that leaves half the oligonucleotide unpaired and the catalytic gap only partly so: they bound the near-match case, and no retrieved measurement bounds the parent case. The two parent compartments of §3.5 sharpen that rather than softening it. For nine designs the route to wild-type *NR4A3* is not a gap-level discrimination problem at all. They pair the catalytic gap in full across the wild-type intron-2/exon-3 boundary, at two mismatches that both fall in the LNA wing, and the compartment in which that duplex would form is the nuclear one RNase-H1 occupies. For 87 the same is true in mature parent sequence, over a contiguous duplex of at least ten base pairs. The general point is that a fusion-junction design's most plausible wild-type liability is its own parent, reached either across a splice junction or in the mature transcript, and both are invisible to a screen that ranks candidates by global identity. The four demonstrations of parental sparing cited here were made in cells, on molecules already synthesised; the comparison above is available before anything is synthesised. Whether other groups apply an equivalent comparison before synthesising is not established here: no survey of published design pipelines was performed, and the argument above is about what particular instruments can see rather than about what the field does. The null of §3.5 is what makes that a finding rather than a restatement of the design rule: arbitrary sequence meets this screen at about 6%, and a chimera keeping the whole rule while joining the two parents at random offsets meets it at 24%, against 46% for designs at real breakpoints. At the ten-base-pair threshold, roughly half the liability is therefore inherent in joining these two transcripts at all, and roughly half is specific to where the disease joins them — a split that holds at that threshold and not across it, the generic share reaching 94% at the loose end of the cited window (§3.5).

Some designs cleave the patient's own un-rearranged *NR4A3* allele, and the parent screen passes every one of them. This is the result most consequential for anyone ordering these oligonucleotides, and it applies wherever the acceptor half of a design is *NR4A3* sequence that is not exonic in the mature transcript: the 5′ untranslated exon 2, and the cryptic exon within intron

2. In a fusion transcript that sequence follows the partner's donor exon. In the patient's

un-rearranged allele the same sequence follows *NR4A3* intronic sequence, so the question is whether the design's donor half also matches that intron closely enough for the whole catalytic gap to pair. For three designs it does. Two are at *EWSR1* exon 13 joined to *NR4A3* exon 2 — 5′-CAGTGGGCTCTCCACG-3′ and 5′-GCAGTGGGCTCTCCAC-3′, each pairing across the wild-type intron-1/exon-2 boundary at two mismatches with neither inside the gap — and one is at *TAF15* exon 6 joined to the intron-2 cryptic exon, 5′-TGATGAGGGCCTTGTG-3′, likewise gap-paired at two mismatches. All three are named here as not to be carried forward, and all three are excluded from every best-design field above.

Naming a design not to use is only half of a usable statement, and at both seams the other half exists. At *EWSR1* exon 13 to *NR4A3* exon 2 the surviving reagent is 5′-AGTGGGCTCTCCACGG-3′, and at *TAF15* exon 6 to the cryptic exon it is 5′-ATGAGGGCCTTGTGTG-3′, whose catalytic gap carries three *TAF15*-derived bases the *NR4A3* locus does not have and which returned no wild-type site at all. Neither seam is therefore lost to this liability; what is lost is the assumption that designs tiled across one seam are interchangeable.

Two things about that finding matter more than the three sequences. The first is how they were reached: each of the three had already cleared the mature-parent exclusion, and so had every other design tiled at its seam, that exclusion being a screen over spliced cDNA and therefore structurally unable to see intronic sequence at all. A clean parent screen at such a seam is the silence of an instrument that cannot look at the compartment in question, and the same three designs were returned independently by an exhaustive scan of the *NR4A3* unspliced sequence, by the pre-mRNA screen and by the genome scan, on a fixed known-positive control that fired on exactly the one design it was required to fire on.

The second is what decides it, because the obvious answer is wrong. It is not the gap-level margin this paper ranks by: the two condemned *EWSR1* designs carry 2 and 1 donor bases inside the catalytic gap, while a design at the same seam with a margin of 1 and five donor bases in its gap is clean. What decides is how much donor sequence the gap holds, because the rest of the gap is acceptor sequence the wild-type allele already carries verbatim. That makes the liability a property of the donor rather than of the acceptor, and the sequence bears it out: *EWSR1* exon 13 ends CACTCCGTGGAG against the last twelve nucleotides of *NR4A3* intron 1, CCTTGCCTGTAG, matching at 7 of 12 positions with a shared terminal AG, whereas *TAF15* exon 6 ends ACCACACACAAG and matches at 4, mismatching in every register — which is why the *TAF15* exon-2 seam returns no such design and the *EWSR1* exon-13 seam returns two. The general rule for this modality is that a design must be checked against the acceptor gene's unspliced sequence whenever its acceptor half is not exonic in the mature transcript, and that no result from a spliced-transcript screen substitutes for it.

All of that presumes that sparing wild-type *NR4A3* is worth the specificity cost, and that premise deserves examination rather than assumption. The published evidence cuts both ways and neither way is decisive. On the permissive side, *NR4A3* has two close paralogues and the family is functionally redundant where it has been tested: NR4A1 and NR4A3 are described as functionally redundant suppressors of acute myeloid leukaemia, and the three receptors are highly homologous.[39,40] A conditional double knockout of *Nr4a1* and *Nr4a3* disturbs haematopoietic stem-cell homeostasis, and even then the cells retain regenerative and differentiation capacity;[39] no single-knockout arm is reported there, so the double deletion is shown to be sufficient and not to be necessary. On the restrictive side, a separate study makes the loss of *NR4A3* consequential rather than silent when paralogue reserve is reduced: mice hypoallelic across the two genes develop a myelodysplastic or myeloproliferative neoplasm, and abrogation of both leads to rapid postnatal leukaemia.[41] The family is also not uniform in direction — within atherosclerosis, NR4A1 and NR4A2 attenuate lesion formation while NR4A3 aggravates it[42] — so paralogue redundancy cannot be assumed to be substitution.

Two limits on that reading matter more than the reading itself. Every study cited here is haematopoietic or vascular, and none addresses the tissue an extraskeletal myxoid chondrosarcoma arises in; and all describe germline or conditional gene deletion, which is a different and more complete perturbation than partial, reversible, dose-limited knockdown by an oligonucleotide. The honest position is therefore that wild-type *NR4A3* knockdown has an unquantified cost that is probably not zero and probably not catastrophic, and that the case for junction selectivity does not rest on it. It rests on the *EWSR1* and *TAF15* side: the fusion's partner genes are essential RNA-binding proteins, and a reagent cleaving a parent transcript is failing at the one thing that distinguishes this modality from knocking down *NR4A3* directly, which requires no junction at all. A design that cannot spare the parents has no advantage left to trade.

Free-energy calculation does not narrow the interval either. Every design discriminates amply at the level of duplex formation, so what is unresolved is specifically the catalytic step, not the binding one. Two things could narrow that interval, and no further sequence analysis is either of them: a measurement, or a physics-based estimate of cleavage geometry on the RNase-H1·heteroduplex complex, for which experimental structures exist. Neither is attempted here. Gap length is not a third, for a reason that is arithmetic rather than empirical. In every design of all three panels, the margin a longer gap wins and the contiguous parent duplex it concedes are the same nucleotides (§3.8). A longer gap buys a markedly quieter transcriptome, and buys it by making RNase-H1 more competent against the parent as well as against the fusion. That is the same limit reached from the other side rather than a way around it. The field's own answer to poor single-base discrimination has been positional chemical modification of the gap rather than length,[32] and that is the design direction this result points to, now for a demonstrated reason rather than by analogy. A steric-block mechanism, which does not require gap-level discrimination, is a second alternative this work does not evaluate.

Delivery remains unsolved for a tumour, and separates into three routes with different requirements. A characterised EMC-enriched surface antigen is a prerequisite of the systemic receptor-targeted route only; local and inhaled administration require none. EMC's distant spread is lung-dominant, at 35–45% of patients and a median of approximately 28 months to metastasis.[6] Inhaled oligonucleotides have reached human dosing in non-oncology indications. An inhaled antisense oligonucleotide has been dosed in healthy volunteers in phase 1,[43] though that was a splice-switching oligonucleotide rather than an RNase-H1-active gapmer, so it establishes the route and not the mechanism used here. An inhaled siRNA has reached phase 2b–3 in patients.[44] Both target airway epithelium or parenchyma, which is the compartment inhalation naturally reaches. A hypocellular, matrix-rich parenchymal sarcoma nodule is not. Inhaled delivery to lung tumours is an active preclinical field, with 68 records in the retrieval corpus behind this section, but only two of those carry clinical-stage language and neither is a trial. The route is therefore established in humans and not for this target.

The testable surface is narrower than the literature makes it look, and a reader planning an experiment should know that before ordering anything. A junction-spanning gapmer needs a junction to span, so the test article has to carry one. The extraskeletal myxoid chondrosarcoma line a reader would reach for first, H-EMC-SS, is the only one this work could establish as available from a cell repository, a third line's distributor being unreadable and its availability therefore unanswered rather than answered, and no *NR4A3* fusion is detectable in it on the public record: a filtered fusion caller that ran against it returned two calls, neither naming *NR4A3* nor any FET gene; its *NR4A3* expression sits at floor; the reference registry that records a gene fusion for other extraskeletal myxoid chondrosarcoma models records none for this one; and no retrieved source reports a positive junction in it. The operative consequence is narrow and is the only one this paper draws: no reagent named here can be tested in that line. This is not a statement that the line is misidentified — it carries a concordant short tandem repeat profile from three independent sources and no problematic-line flag — nor a statement about what the line is instead, and fusion-negative extraskeletal myxoid chondrosarcoma tumours are themselves a recognised minority category, so absence of the fusion is not by itself a reclassification. The observation is also not new: it is in print in one figure legend and is carried as a caution field in the reference registry. No retrieved source examines it as a subject, and it is not discoverable by anyone searching on model validity.

What remains is five test articles, and each of the five now has a matching reagent. Three are the engineered constructs of the functional study cited above,[45] E-N, T-N* and T-N, whose exon spans that paper states verbatim; two of the three are junctions this work screened at full panel depth and the third is the intron-2 cryptic-exon seam of §5.1. The other two are the patient-derived, identity-clean models reported with two extraskeletal myxoid chondrosarcoma tumours,[46] whose fusions are reported as *EWSR1* exon 13 and *TAF15* exon 6 joined to *NR4A3* exon 2 rather than exon 3; reagents exist at both acceptors, so each line has one under either reading of that exon label.

The two routes are not interchangeable and their limits run in opposite directions. Rebuilding the constructs is the faster route and the only one whose critical path contains no third party who can decline or not reply, since the junction is specified by construction. What it cannot buy is biological relevance: a complementary DNA over-expressed in a heterologous background is not the disease, so such an experiment could speak to junction-selective knockdown of the intended transcript and not to activity at endogenous expression from an endogenous locus. The published recipient background is in no cell-line registry either, so a rebuild would sit in a different background from the original — an isogenic mismatch to declare rather than gloss. The patient-derived models are the only route to a fusion-positive extraskeletal myxoid chondrosarcoma cell, and are available on request from the originating laboratory with no repository deposit; what a transfer requires is stated nowhere that could be read, which is an absent statement rather than an absence of conditions, and the cells are slow once received, at reported doubling times of five to six days as sarco-spheres passaged every two to three weeks, which constrains any exposure window. A third reported line cannot serve as a test article at all on current evidence, because its fusion partner and exons are unstated anywhere readable and it would have to be sequenced first. One constraint sits above all of them and no reagent choice moves it: every route ends at someone culturing cells, and this work has no laboratory, so the rate-limiting step is a laboratory rather than a line, a construct or an oligonucleotide.

5 · Reagents, controls and the decisive experiment

This section is the paper's output for a laboratory. It names five things: the oligonucleotides to make, the arm that separates the two ways a weak result could arise, the predicted off-target load each carries, the controls without which the readout does not mean what it appears to mean, and the number that would falsify the ranking every candidate here is ordered by. Nothing in it is a claim of efficacy. No sequence named below has been synthesised or tested.

5.1 · The reagents to synthesise

The experiment that would resolve the central uncertainty has been published in an analogous disease. Fusion-specific antisense oligonucleotides against *NAB2::STAT6* in solitary fibrous tumour, evaluated against CRISPR-engineered isogenic fusion-positive and fusion-negative cells, reduced fusion expression by 58% and proliferation by 22% in vitro.[47]

Applied here, the reagents to synthesise are the best available at the two most frequently reported junctions with a published exon-resolved breakpoint (Table 4): 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6. Both hold the top gap-level margin of 3, and neither pairs a parent through the catalytic gap at the ten-base-pair threshold, although the *TAF15* reagent's longest parent run is nine. The first also tests the multi-partner prediction, against a synthetic target only.

How much of the disease those two junctions represent has to be stated as a junction figure and not as a partner figure, and the two differ substantially. Partner prevalence across 58 molecularly confirmed cases is 46 *EWSR1* and 9 *TAF15*, or 95% between them.[9] These reagents are not partner-specific: each spans one exon pair, and §5.4 requires the breakpoint of any test material to be established at nucleotide resolution before either is ordered. Discounting by the published breakpoint distribution — *EWSR1* exon 12 to *NR4A3* exon 3 in 10 of 15 *EWSR1*-rearranged tumours, and *TAF15* exon 6 to *NR4A3* exon 3 in all three *TAF15*-rearranged tumours of the same 18-case series[37] — gives 68.4%, or roughly two thirds. The interval is wide and the reason is the denominators rather than the estimate: taking each breakpoint fraction to its own Wilson bound spans 39.9% to 82.8%, because the *EWSR1* arm rests on 15 tumours and the *TAF15* arm on three. It also assumes that the breakpoint distribution within *EWSR1*-rearranged tumours is the same in the 58-case cohort as in the 18-case one, which nothing here tests and which no published series is large enough to settle. Two thirds is therefore the honest figure to plan a reagent set around, and the quantity that would sharpen it is breakpoint sequencing of archival material rather than any further analysis of sequence.

That two thirds is a property of the panel size, not of the design method, and the distinction decides what would raise it. No oligonucleotide can serve two breakpoints of the same partner: the donor exons do not resemble each other at the seam, *EWSR1* exon 12 ending AATGGTTTGATG against exon 13's CACTCCGTGGAG, which agree over a single terminal base. Taken across every pair of in-frame junctions of one partner in this panel, the longest shared 3′ donor run is five nucleotides, at *TFG* exons 2 and 6, and three within *EWSR1*. The cross-partner coverage of §3.2 is the exception that shows the rule, and it needs the ten identical donor bases the FET paralogues share. Coverage therefore rises only by adding reagents, and the ladder is short. A third reagent at *EWSR1* exon 13 to *NR4A3* exon 3 — the second most frequent *EWSR1* transcript in the same 18-case series, and one of five cases in an independent whole-transcriptome cohort[48] — takes the set from 68.4% to 79.0%, and its best design, 5′-GGGCATATCTCCACGG-3′, holds the same top gap-level margin of 3. Beyond that the arithmetic runs out before the target does: a set covering every *EWSR1* and every *TAF15* breakpoint reaches 94.8% of molecularly confirmed cases and stops, so no panel restricted to those two partners reaches 95%, and the remaining reachable cases are the two *TCF12* tumours. A fourth reagent, 5′-GGGCATATCCATCAGA-3′ at *TCF12* exon 5 to *NR4A3* exon 3, addresses them. Its junction is resolved to the nucleotide by the deposited chimeric cDNA of §3.3, it holds the same top gap-level margin of 3 with no parent liability, and it has been through the same five screens; its transcriptome load is 17 gap-paired near-matches at a single locus. What it does not come with is a distribution. One *TCF12*-rearranged tumour has ever been sequenced at this junction, and neither breakpoint series used above contains a *TCF12* tumour at all, so the arm can only be priced at its ceiling and the resulting 98.3% is an upper bound rather than a reachable target.

The *EWSR1* exon-13 reagent should not be recommended on its transcriptome count, because the two axes that separate it from the exon-12 reagent point in opposite directions and only one of them bears on where an effect would land. On count it is the lighter of the two, 24 gap-paired near-matches at 2 loci against 123 at 6. On exposure it is the heavier: both of its loci are transcribed at the upper cut in the organs a systemic phosphorothioate gapmer distributes to, while none of the exon-12 reagent's four measurable loci reaches that cut in liver or either kidney compartment, its largest sitting instead in brain and gut (§3.7, Table 6). For a laboratory choosing between them, the exposure reading is the one that speaks to the question a count cannot: a locus matched but not transcribed where the drug goes has no route to an effect, whereas the count is a property of how densely the returned loci are annotated. Neither axis is a risk ranking, and this comparison does not make the exon-12 reagent the safer molecule. Every hit behind both is a 14 of 16 match, no cleavage is predicted at any of them, and an expressed gene is necessary and not sufficient for an effect. The reason to make the exon-13 reagent is coverage, and it is unaffected by either axis.

Nine junctions now carry both a published exon-resolved breakpoint and a design through all five screens, and four of them sit outside the 38-junction panel, at the *NR4A3* exon-2 acceptors named below. Four of the nine move the estimate; one moves only the bound, its arm having no measured within-partner distribution; and four move it by nothing at all — two because their partner is absent from this cohort, and two because their partner is present while their exon pair carries no count in it. That last pair is the easier of the two zeroes to miss, and a membership test that asked only about the partner did miss it. Priced on a pooled breakpoint basis rather than on the single series the ladder uses, the nine together are 82.9% of molecularly confirmed cases, widening to 57.5–90.7%. The denominators are not the ladder's: the *EWSR1* arm moves from 10 of 15 to 17 of 20 by pooling the 18-case series with the five-case whole-transcriptome cohort, while the *TAF15* arm stays at 3 of 3. Three things follow. Fifteen of those 20 tumours come from one series, so the pool stays close to it; the two series agree at *EWSR1* exons 12 and 13 and disagree completely at exon 7 to *NR4A3* exon 2, 0 of 15 against 1 of 5; and a third series was refused because every case in its *EWSR1* arm carries a covered junction by construction, which would fix its rate at 100% and raise the figure rather than lower it. Two of seven retrieved series resolve every case to an exon pair, so this is a pooled record and not the whole one. The figure supersedes nothing: it answers a different question from the ladder, whose rungs are incremental and priced on the single series 68.4% is computed on, and 68.4% remains the coverage of the two reagents named at the head of this section. Its own membership rule is an evidence test rather than a list — a junction qualifies where a published report places a patient's breakpoint at it and a reagent has been through all five screens — and one qualifying junction, *PGR* exon 2 to *NR4A3* exon 2, reported in a single patient,[49] moves the figure by exactly zero, because the 58-case cohort behind the denominator contains no *PGR* case for such a reagent to engage. *PGR* is a sixth partner, outside the five modelled here, and §2 gives no transcript accession for it, so the reagent named at that seam below is screened against the non-canonical-acceptor table rather than derived through the panel's own transcript models. *TFG* exon 7 to *NR4A3* exon 3 is the second, on the same ground and from a deposited cDNA rather than a report. What that reagent changes is which patients are reachable at all, which is a different statement and is not added to a coverage percentage.

One junction in that ladder cannot be built from this panel at all, and the obstacle is a modelling choice rather than a sequence. Every junction screened here joins a donor exon to *NR4A3* exon 3, because the atlas that emits them drops exon-2 acceptors as non-coding — *NR4A3* exon 2 lies upstream of the start codon. The *EWSR1* type 2 transcript joins *EWSR1* exon 7 to *NR4A3* exon 2,[50] and it is not hypothetical: it was sequenced as one of the five cases of that cohort[48] and again in an independent patient.[51] A frame-based exclusion is correct for a fusion protein and does not transfer to an RNase-H mechanism, which cleaves a transcript whether or not its reading frame survives. The same applies to the *TAF15* exon 6 to *NR4A3* intron 2 variant used in functional work.[45] Emitting and screening those acceptors was the largest piece of designable but undesigned coverage this analysis identified, and the two are no longer in the same state.

The *NR4A3* exon-2 acceptors are now designed and screened to the panel's depth, at four seams with a published exon-resolved breakpoint, each tiled by the same five registers and graded by the same five screens. Their best available designs are 5′-CAGTGGGCTTCTGCTG-3′ at *EWSR1* exon 7, the type 2 transcript, at gap-level margin 2 and 51 gap-paired near-matches over 7 loci; 5′-AGTGGGCTCTCCACGG-3′ at *EWSR1* exon 13, at margin 3 and 25 over 6; 5′-AGTGGGCTCTTGTGTG-3′ at *TAF15* exon 6, at margin 3 and 128 over 6; and 5′-AGTGGGCTCTTCCATT-3′ at *PGR* exon 2, at margin 3 and 51 over 14. That last seam carries a caveat the other three do not: *PGR* is a sixth partner, outside the five modelled here, and §2 lists no transcript accession for it, so its design is screened against the non-canonical-acceptor table rather than derived through the panel's own transcript models. None of the four is clean, and the least loaded of them is the least dirty of five dirty designs rather than a different kind of result. They are reported beside the panel and never pooled into it, because the grade that excludes their junctions from the 38 is unchanged: the manuscript's panel is 38 junctions and these are not among them. Two further designs at the *EWSR1* exon-13 seam carry a liability the parent screen cannot see and are excluded for it, described in §4.

The *TAF15* intron-2 acceptor and a second cryptic-exon seam, *EWSR1* exon 10 to the same acceptor, are designed but not comparably screened: three of the five instruments cannot address them at all, because they resolve a junction by exon index and a cryptic exon 5′ of the *NR4A3* start codon has none. Their counts are therefore absent rather than low, and must not be read beside the panel's. What those two seams do carry is the pre-mRNA and genome arms, which are the two that interrogate the compartment an intronic acceptor lives in, and at the *TAF15* seam those arms decide between the five designs rather than ranking them (§4).

Two risks attach, in this order. The first is architectural, and the Methods disclose it. A six-nucleotide gap supports noteworthy but incomplete RNase-H1 activity where seven to ten are reported as optimal,[25] so weak knockdown is at least as likely to be the gap as the sequence. That risk is now addressable by a named second reagent rather than by a caveat.

5.2 · A second geometry as a gap-length control

5′-AGGGCATATCATCAAACC-3′ is the 5-8-5 design at the same *EWSR1* exon 12 junction. It spans the same three partners' breakpoints and sits inside the reported activity optimum. It holds a gap-level margin of 4 where the 16-mer holds 3, and carries 3 sense-strand near-matches across the gap at one gene locus, against the 16-mer's 123 at six (§3.8, Table 5).

Synthesised alongside the 16-mer, at one extra oligonucleotide and one extra well per condition, it separates the two explanations a weak result would otherwise confound. A 5-8-5 arm that knocks down where the 5-6-5 arm does not attributes the failure to gap length rather than to sequence. What it does not buy is parental sparing, since the same two nucleotides lengthen each parent's contiguous duplex from 3 to 4 nucleotides of gap DNA, its whole contiguous hybrid at that seam from 8 to 9 base pairs, and its free energy from −7.77 to −8.66 kcal/mol. Its longest mature-parent duplex through the whole gap does fall from 8 base pairs to none, but 8 sits below the ten-base-pair threshold applied throughout, so neither design counted as a mature-parent liability at this seam and that fall removes nothing the screens had counted. Both arms therefore need the fusion-negative comparator below.

5.3 · The predicted off-target load of each reagent

The second risk is transcriptome load, and it differs sharply between the two reagents. The *EWSR1* reagent carries the heavier load of the two named here: 123 gap-paired sense-strand near-matches at the deeper ceiling, recounting to six gene loci, all at the screen's loosest admitted identity and none on a parent transcript (§3.10). It is not the heaviest in the panel — fourteen design records carry more, to a maximum of 240, and Table 2 prints them. The *TAF15* reagent carries 8 such near-matches at five loci.

The parent compartments qualify that, in a way the transcript screens cannot show. The *EWSR1* reagent carries a sense-strand intron–exon-spanning near-match in wild-type *TAF15* pre-mRNA at two mismatches, one of them inside the catalytic gap, returned independently by the pre-mRNA screen and the genome scan. It falls outside every parent count reported here, because those require the gap to be paired in full, and by the bounds adopted above a single gap mismatch does not abolish cleavage. It is the multi-partner result's own cost rather than an incidental hit: the ten donor bases shared across *EWSR1*, *TAF15* and *FUS* that let one oligonucleotide span three junctions are the bases that place it against wild-type *TAF15*. The *TAF15* exon-6 reagent carries no sense-strand pre-mRNA site at all, which is a second respect in which the two separate on something other than count.

That load should travel with the reagent. It is a liability to disclose and to control for rather than a disqualification, because on the genome scan the same design falls below chance in both directions that matter: 0.69 times the expected number of near-matches at two mismatches, and 0.62 times the expected number of gap-paired ones. Expression reads the two loads differently from their sizes (§3.7). None of the *EWSR1* reagent's four measurable loci is expressed at the upper cut in the organs a systemic dose reaches, while the *TAF15* reagent's five include *NRP1*, which is expressed at that level in all three. That does not reverse the ranking, since no screen here establishes that a two-mismatch duplex engages any of them, and it is not a statement about safety. It is the first evidence separating the two reagents on anything but count, and it points the load question at the shorter list rather than the longer.

5.4 · Controls and the decision threshold

The three designs that survive every screen are mechanism controls rather than candidates: 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8, 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 and 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7, tiered as §3.6 describes. The *TAF15* exon-1 design is contradicted by the exon-6 breakpoints reported in EMC, and the *TCF12* exon-7 design is contradicted in the same way by the exon-5 junction of §3.3; for *FUS* no exon-resolved EMC breakpoint has been published at all. 5′-GGGCATATCTCTATAA-3′ at *TCF12* exon 17, which an earlier draft of this work put forward for that role, is not among them. At ten times the default search depth it carries 101 sense-strand near-matches, 14 of them spanning the catalytic gap.

Transferability depends on how that experiment is set up. The breakpoint of the cell line or patient sample must be established at nucleotide resolution by RNA sequencing before any oligonucleotide is ordered: every design here is specific to one exon pair, and none is valid for an unverified junction. Routine diagnosis does not supply it: break-apart *NR4A3* fluorescence in situ hybridisation is the preferred single assay because it detects any rearrangement "irrespective of partner",[6] so on its own it does not locate the seam, and the 58-case cohort every coverage figure here is denominated on was identified that way, with RNA exome sequencing applied to three of its cases.[9] That cohort reports a partner for 57 of its 58 tumours, so partner assignment there rests on more than the break-apart assay; what no part of that workflow supplies is the exon pair.

Three controls are required, and a knockdown assay alone distinguishes none of them:

- a positive control gapmer against an abundant housekeeping transcript in the same cells, to separate failed delivery from failed discrimination;
- a scrambled gapmer of matched chemistry, to separate sequence-specific cleavage from the non-specific toxicity of this chemistry;

- a fusion-negative isogenic comparator, since wild-type *NR4A3* may be too weakly expressed in an EMC line for the selectivity readout to be defined at all.

Two limits of that set are invisible within it. The normalising reference transcript must be neither the positive control's target nor either transcript measured, or the positive-control well becomes uninterpretable and every other well is rescaled against a perturbed reference. And none of the three separates RNase-H1-dependent cleavage from steric block, which §4 names as unevaluated, so a knockdown seen with this set is consistent with both while the ranking under test models cleavage alone.

Where the wild-type measurement sits decides the answer, because the fusion carries *NR4A3* exons 3 to 8 and a wild-type assay must not read sequence the fusion also carries. The two available placements bias in opposite directions. An amplicon upstream of the acceptor cannot detect the fusion, but it lies on the 5′ fragment that survives cleavage, so it under-reads wild-type knockdown and inflates apparent selectivity; an amplicon spanning the wild-type exon-2/exon-3 boundary reads knockdown directly, but at the *NR4A3* exon-2 acceptors of §5.1 the fusion itself carries that boundary. Both are defensible, no assay is prescribed here, and which placement was used should be reported with the result.

The decision threshold should be fixed before the experiment, in a form that can be registered. Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose–response in the same wells, so that a larger number is a more selective reagent. The cut is defined for that quantity and no other. A ratio of residual transcript at a single dose is not commensurate with it and must not be compared against the same cut: that ratio is bounded above by one divided by the fusion knockdown's complement, so at the 58% knockdown reported for the analogous published experiment[47] it cannot exceed about 2.4 however selective the reagent is, and a cut of 5 would return falsification as arithmetic rather than as biology.

The replicate count follows from the variance rather than being asserted. Selectivity here compounds four normalised measurements, so the replicate standard deviation of its logarithm is the quantity to estimate in a pilot. At a standard deviation of 0.35 on that scale, six independent biological replicates give about 80% power to falsify a true selectivity of 3, and three give about 30%; above a standard deviation of about 0.65 no observed ratio can place a 95% upper bound below 5 at three replicates, so the test cannot fail at all and the design is void rather than negative. The number of replicates should therefore be set from the pilot estimate, with three as a floor and not a target, and the ranking is falsified only where the upper bound of the 95% confidence interval lies below the cut, never where a point estimate does. The ratio is reportable only where the wild-type *NR4A3* knockdown is itself resolved above a pre-stated limit of quantification — the quantity that can approach zero, which is the change in wild-type transcript and not its vehicle-well abundance; a wild-type change indistinguishable from none returns an unbounded selectivity that would otherwise pass by default, and the honest output there is a one-sided lower bound, which cannot falsify. The cut is 5.0, taken as a convention from the approximately five-fold near-match figure of §2 rather than measured for this comparison, on which §4 records that no retrieved measurement bounds the parent case. It sits at the optimistic end of the one- to five-fold span reported above, so a reading inside that span is consistent both with the ranking failing and with the discrimination this chemistry is already reported to give; the margin contrast below is what separates those two, and a single-arm result does not.

The reagents of §5.1 all hold gap-level margin 3, so as a set they measure the level of selectivity at the top margin rather than whether selectivity orders by margin. One optional arm supplies that contrast, and only at *EWSR1* exon 12: 5′-GCATATCATCAAACCA-3′ is the margin-1 register of the same junction at the same geometry. Margin is not the only variable that moves with the register, and the others are named here rather than left for a reader to find: GC falls from 43.8% to 37.5%, outside the 40–60% window §3.9 audits, so this arm would fail a conventional rule the lead reagent passes; gap-paired near-matches fall from 123 to 34 while single-mismatch off-targets rise from 1 to 22; and its longest mature-parent duplex is eight base pairs against wild-type *FUS*, which is **the same length the lead**

reagent carries against wild-type *TFG*, so the two do not separate on that screen at all. A difference between the arms is therefore not attributable to margin alone.

One asymmetry decides what the arm can show. Its five parent-paired gap nucleotides are *EWSR1*, the donor, not *NR4A3* — so the liability its low margin creates is invisible to a readout defined on wild-type *NR4A3*, and interpreting this arm requires the wild-type *EWSR1* measurement alongside it. The mirror register at the same seam, 5′-CAGGGCATATCATCAA-3′, is the one whose five fall on *NR4A3*, and it carries an eleven-base-pair duplex against wild-type *NR4A3* that would confound it differently.

TAF15 exon 6 cannot supply the same arm: all four of its lower-margin registers pair a parent through the whole gap at eleven or twelve base pairs, two of them against wild-type *NR4A3*. A contrast at one junction tests the ordering at that seam and not the ranking across the panel, and the comparison it supports is between two arms rather than of one ratio against the cut, for which this section states no rule.

6 · Limitations

Search depth. The cleanliness claim is bounded by what each screen can see. The alignment screen returned hit lists for 183 filtered designs, and only 47 of those 183 are short enough to be assessed for cleanliness at all, so nine clean designs is a floor over that subset rather than a total — and simultaneously an over-count, for the censoring reasons measured in §3.4 and bounded in §3.10. The default-depth figures throughout should be read as lower bounds rather than as counts. BLAST is also heuristic, and its sensitivity at the ≥14/16 threshold is unquantified here, so "no sense-strand near-match" is a property of this search rather than of the transcriptome. The exhaustive transcript scan carries no such qualification, and is the screen the claim rests on. All 38 in-frame junctions are screened with the orientation filter applied, so no junction here carries an unfiltered count. The chance null is crude: it assumes independent uniform bases, where real transcript sequence is composition-skewed and repetitive, so it separates "more than chance" from "at chance" and nothing finer.

The *TAF15* arm's coverage is an upper bound, permanently. Every coverage figure here prices the *TAF15* arm at 3 of 3 — every *TAF15*-rearranged tumour joining exon 6 to *NR4A3* exon 3 — on a three-tumour series. A functional study reports a second isoform of the same fusion, joining *TAF15* exon 6 to a cryptic exon within *NR4A3* intron 2, and calls the two "the two major *TAF15-NR4A3* isoforms detected in human tumors" while ordering them only as commoner and less common.[45] No count of either has been published, and the number that would settle it is not one this work can obtain, so any patient carrying the intron-2 isoform is outside the coverage figures rather than inside them, and 68.4% is optimistic on that arm by an unmeasured amount. Three measured statements bound the consequence. First, no reagent in this panel engages that isoform: 0 of the 190 designs place a catalytic gap across its seam. Second, the two isoforms share no sequence 3′ of the breakpoint, so this is not a near-miss that a longer or shifted oligonucleotide recovers — it is a different target. Third, a design at that seam now exists, emitted against a cryptic exon measured from the genome rather than assumed, so the gap is a retrieval gap and not a design gap.

***TCF12* recurrence is untested, not refuted.** The *TCF12* arm rests on one sequenced tumour, and the 98.3% figure is an upper bound for that reason. It should not be read as evidence that the junction is non-recurrent. *TCF12*-rearranged extraskeletal myxoid chondrosarcoma has been counted in at least four series and sequenced once, in 2000; the later cohorts identified their cases by break-apart fluorescence in situ hybridisation, which reports that the *NR4A3* locus is split without locating the seam within it. A junction that no assay in those cohorts could have resolved cannot be absent from their results in any informative sense. The distribution is therefore unmeasured rather than measured to be narrow, and it would take sequencing of archival material to move it.

Which junction a patient actually carries. Which exon pair a given patient carries is not decidable from exon structure. The multi-partner result is therefore conditional on *TAF15* and *FUS* breakpoints falling at the homologous exons, a clinical fact not established here. The *TCF12* exon assignment of §3.3 rests on a deposited sequence rather than on exon structure, and one tumour has been sequenced at that junction, so nothing here says how a second *TCF12*-rearranged tumour breaks. The five partners modelled here are also not the full catalogue: *ACTB*[3] and others are reported, and 2% of one cohort carried no identified partner.[9]

Some reported breakpoints have no exon pair to name them by. Three of the 15 *EWSR1*-rearranged tumours of the primary breakpoint series carry transcript types the retrieved record does not name, and that block is the largest single distance between the reagent set above and the arithmetic ceiling. It is usually written up as a retrieval problem, and retrieval is an upper bound on what would open it. At least one named variant of this fusion is reported to arise from a genomic breakpoint interior to *EWSR1* exon 12 rather than between two exons, in a source that carries a citation marker on that sentence and is therefore restating an earlier report.[52] Such a breakpoint is not undesignable: the chimeric transcript it produces still has one definite nucleotide seam, an RNase-H1 gapmer cleaves a transcript rather than an exon boundary, and handing the same builder a donor model cut inside its exon returns five candidates, all fusion-specific against both parents, three gap-centred and a best gap-level margin of 3 — the panel's own geometry. What such a junction lacks is an exon index, which is how every design here is specified, and a published nucleotide position, which no retrieved source states. It is therefore out of reach for a stocked panel while remaining designable for a named patient whose breakpoint has been sequenced. How many tumours this accounts for is not established by any source, and no estimate of it is offered here.

Geometry. Every screened count outside §3.8 is for one architecture, a 16-mer 5-6-5, and §3.8's comparison carries bounds of its own. The genome scan is unavailable at 18 and 20 nucleotides by construction rather than merely unrun, since the scanner is a packed bitmap over the code space of a 16-mer and a longer window needs a different data structure. The nesting argument bounds a longer design's genome liability by that of its own sub-windows, but no such scan has been run, so that bound is an available next step and not a result. No RNase-H1 assay distinguishes these geometries here, so which gap is preferable is not decided by this work.

What the screens do and do not model. The thermodynamic calculation models an unmodified DNA:RNA hybrid, and speaks to duplex formation rather than to cleavage. Every parent count reported here requires the catalytic gap to be paired in full. That is the binary rule the Methods decline to apply to cleavage, used here as an inclusion criterion because no retrieved measurement grades a partly-paired parent duplex; the class it excludes is 21 designs at the pre-mRNA screen, and is stated in §3.5 rather than left to the artefacts. All five screens address hybridisation-dependent liability only. The sequence-independent liabilities of a phosphorothioate LNA gapmer, protein binding and the target-independent hepatotoxicity of this chemistry, are not a function of any feature graded here.

Expression. The expression reading carries bounds of its own. Thirteen of the 46 loci returned no reading, and they carry 52 of the panel's 649 records, so for those the exposure question is unanswered rather than answered negatively. Several are attributable to what the locus is: a brain-associated long non-coding RNA host, an antisense transcript, a readthrough, and uncharacterised loci carrying only predicted transcript models. No expression figure is a predicted cleavage event, and the step from a gene being expressed in liver to that oligonucleotide being a problem needs an argument no screen here supplies (Table 6). Reference bulk medians describe normal tissue in a population, not a dosed patient's organ.

The genome scan. Screen 5 removes the six-transcript bound the parent screens carried, at the cost of two bounds of its own. Whether a site lies on a gene's sense strand is decided against an annotation, so a site in unannotated transcription is reported as intergenic and not counted. And a screen against one assembly says nothing about a patient's private variation. An earlier genome-wide attempt against a mixed public corpus returned nothing interpretable and is released with the artefacts; it could not have done otherwise, having no defined nucleotide span to form a null against. Every exhaustive screen also inherits the substitution-only bound: all are complete for mismatches by construction and blind to insertions and deletions.

Tables

Table 1. The in-frame junction space across five *NR4A3* fusion partners. Every donor-exon × *NR4A3*-acceptor-exon pair was graded against the frame condition before any design was emitted. The gap-level margin is the number of junction-unique bases inside the six-nucleotide catalytic gap on the shorter side of the junction. Frame compatibility is an arithmetic property of exon structure and is not a claim about which junctions patients carry.

5′ partner	donor exons	exon pairs graded	in-frame	with ≥1 fusion-specific design	GC range of those designs (%)	best gap-level margin
EWSR1	17	51	8	8	37.5–75.0	3
FUS	15	45	8	8	31.2–62.5	3
TAF15	16	48	8	8	31.2–68.8	3
TCF12	21	63	8	8	25.0–56.2	3
TFG	8	24	6	6	25.0–50.0	3
all 5 partners	—	**231**	**38**	**38**	—	—

Table 2. Predicted specificity per screened junction. One row per junction; figures are for the design with the highest gap-level margin at that junction, which is the ranking the Methods define, and NOT for that junction's cleanest design — the two are often different molecules, and the cleanest ones are in Table 3. Near-match counts are of RefSeq transcript accessions and are also given collapsed to distinct gene loci, since RefSeq carries one accession per annotated variant. A “≥” marks a right-censored count: the screens store the top 15 hits per design, so a design with more is a lower bound. All 38 junction screens are filtered by alignment orientation. `XM_`/`XR_` records are computationally predicted gene models rather than curated transcripts, and are counted separately for that reason. None of these numbers is a measurement of off-target activity.

[1] Counted over the gap-spanning loci only, not over all of that design's near-match loci.

[2] A near-match count is what the search returned on EITHER strand; a match on the strand opposite the target window cannot be hybridised by an antisense oligonucleotide and is not a liability. Across this corpus 44% of apparent gap-spanning hits (738 of 1,677) are of that kind, which is why the two columns differ and why the raw count alone should not be read as load. This column counts only the 15 RETAINED hits. The gap-spanning locus column is recounted from those hits wherever they are the complete list, and is exact there; a “≤” marks a truncated design, where the column instead carries the screen's own count over every ranked hit, computed under a locus assignment since corrected that split some genes across accessions and therefore over-counts. The two columns are not in conflict where a truncated design shows “≥0” sense-strand hits and a non-zero gap-spanning locus count: the sense-strand hits are real and simply fall outside the stored window, which is precisely why such a design cannot be called clean.

[5] The same design re-screened at a tenfold deeper alignment ceiling, with retention raised to match it so that no hit list is truncated. Because no list is truncated, the gap-spanning locus column at this depth is recounted from the complete stored hits under the current locus assignment and is exact; it is not the screen's own stored figure, which was computed before that assignment was corrected and splits any gene whose description carries a comma across one accession per transcript variant. It is therefore the same quantity, counted the same way, as the locus figures in Table 4 and in the Results. The three columns are the counterparts of the default-depth columns to their left, given beside them rather than in place

of them because the default depth is where the corpus-wide counts elsewhere in the paper were computed and the two must stay comparable. Read together they are the paper's censoring result at the level of a single row: a default-depth count is a lower bound whether or not it reached the 50-hit cap, and three junctions whose default cell reads zero in the gap-spanning column carry gap-spanning hits at ten times the depth. A “—” means the deeper re-screen returned no result for that design and is not a count of zero; three of the panel's 190 records failed at this ceiling.

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand[2]	loci with a gap-spanning hit	of those, predicted models only[1]	at the deeper ceiling: near-matches[5]	of those, on the sense strand[5]	loci with a gap-spanning hit[5]	≤1-mismatch matches across that junction's designs, median (max)
EWSR1 e10::NR4A3 e3	5	3	5′-GGGCATATCTAGATCA-3′	35 → ≥2	≥15	≤4	0	138	137	5	4 (32)
EWSR1 e12::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
EWSR1 e13::NR4A3 e3	5	3	5′-GGGCATATCTCCACGG-3′	31 → ≥3	≥12	≤2	0	63	47	2	2 (25)
EWSR1 e15::NR4A3 e3	5	3	5′-GGGCATATCCGGGGGC-3′	36 → ≥2	≥12	≤1	0	93	62	1	1 (10)
EWSR1 e1::NR4A3 e3	5	3	5′-GGGCATATCCGTGGAC-3′	0 → 0	0	0	0	27	18	0	0 (0)
EWSR1 e4::NR4A3 e3	5	3	5′-GGGCATATCAGTGGGA-3′	11 → 4	11	3	2	90	67	8	4 (10)
EWSR1 e7::NR4A3 e3	5	3	5′-GGGCATATTCTGCTGC-3′	32 → ≥9	≥6	≤2	0	300	65	6	9 (21)
EWSR1 e9::NR4A3 e3	2	3	5′-GGGCATATCACCAGGC-3′	29 → ≥2	≥6	≤2	0	165	81	7	1 (22)
FUS e10::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
FUS e11::NR4A3 e3	5	3	5′-GGGCATATCTCCTCGC-3′	30 → ≥1	≥15	≤1	0	60	30	1	0 (3)
FUS e13::NR4A3 e3	5	3	5′-GGGCATATCCATGTGA-3′	6 → 5	2	2	1	20	8	2	1 (9)
FUS e1::NR4A3 e3	5	3	5′-GGGCATATCGTTTGAG-3′	≥50 → ≥7	≥13	≤1	0	129	116	2	10 (38)

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand²	loci with a gap-spanning hit	of those, predicted models only¹	at the deeper ceiling: near-matches⁵	of those, on the sense strand⁵	loci with a gap-spanning hit⁵	≤1-mismatch matches across that junction's designs, median (max)
FUS e3::NR4A3 e3	4	3	5′-GGGCATATTGTTCTGG-3′	18 → ≥4	≥1	≤2	0	148	34	4	3 (23)
FUS e5::NR4A3 e3	5	3	5′-GGGCATATCTCCACCT-3′	32 → ≥8	≥8	≤2	1	127	47	4	2 (3)
FUS e7::NR4A3 e3	5	3	5′-GGGCATATCACCAAAT-3′	34 → ≥5	≥8	≤0	0	141	107	4	12 (39)
FUS e8::NR4A3 e3	5	3	5′-GGGCATATCGGAGTCA-3′	4 → 3	3	1	0	4	3	1	0 (1)
TAF15 e11::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
TAF15 e12::NR4A3 e3	4	2	5′-AGGGCATATCTCGCCG-3′	42 → ≥4	≥11	≤3	0	42	33	3	3 (4)
TAF15 e14::NR4A3 e3	5	3	5′-GGGCATATCTCCTCCT-3′	≥50 → ≥2	≥11	≤2	0	174	95	6	19 (40)
TAF15 e1::NR4A3 e3	5	3	5′-GGGCATATCCGACATG-3′	5 → 2	0	0	0	5	0	0	0 (0)
TAF15 e4::NR4A3 e3	5	3	5′-GGGCATATCTGACTGA-3′	34 → ≥3	≥15	≤4	1	78	57	7	10 (53)
TAF15 e6::NR4A3 e3	5	3	5′-GGGCATATCTTGTGTG-3′	11 → 7	7	4	2	62	10	5	2 (2)
TAF15 e8::NR4A3 e3	5	3	5′-GGGCATATCACCAAAA-3′	41 → ≥4	≥13	≤2	0	133	93	3	10 (17)
TAF15 e9::NR4A3 e3	5	3	5′-GGGCATATCAGCATCT-3′	23 → ≥7	≥0	≤2	0	68	48	5	1 (4)
TCF12 e11::NR4A3 e3	5	3	5′-GGGCATATCTAGAATG-3′	15 → 5	2	1	1	46	17	4	2 (6)
TCF12 e13::NR4A3 e3	5	3	5′-GGGCATATCTGTGAGA-3′	17 → ≥8	≥6	≤1	1	170	80	4	1 (12)
TCF12 e17::NR4A3 e3	5	3	5′-GGGCATATCTCTATAA-3′	8 → 3	0	0	0	118	101	5	1 (6)
TCF12 e19::NR4A3 e3	5	3	5′-GGGCATATCTCTGACT-3′	12 → 4	8	1	1	102	79	3	1 (2)

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand[2]	loci with a gap-spanning hit	of those, predicted models only[1]	at the deeper ceiling: near-matches[5]	of those, on the sense strand[5]	loci with a gap-spanning hit[5]	≤1-mismatch matches across that junction's designs, median (max)
TCF12 e3::NR4A3 e3	5	3	5′-GGGCATATCTGATCCA-3′	15 → 4	10	2	2	374	246	6	2 (35)
TCF12 e5::NR4A3 e3	5	3	5′-GGGCATATCCATCAGA-3′	26 → ≥1	≥15	≤17	0	83	70	1	2 (32)
TCF12 e7::NR4A3 e3	4	2	5′-GGCATATCAAGCGCTG-3′	2 → 2	0	0	0	2	0	0	0 (1)
TCF12 e9::NR4A3 e3	4	3	5′-GGGCATATCTTGCATA-3′	14 → 2	8	1	0	86	64	6	8 (23)
TFG e2::NR4A3 e3	5	3	5′-GGGCATATCTTCATCT-3′	≥50 → ≥1	≥15	≤2	0	207	117	5	35 (87)
TFG e3::NR4A3 e3	5	3	5′-GGGCATATCAAATAAT-3′	14 → 6	9	3	0	217	161	6	9 (17)
TFG e4::NR4A3 e3	5	3	5′-GGGCATATCATTTTCA-3′	≥50 → ≥3	≥15	≤5	2	318	238	15	41 (100)
TFG e5::NR4A3 e3	5	3	5′-GGGCATATCTGAAACC-3′	41 → ≥6	≥11	≤7	1	112	72	10	3 (21)
TFG e6::NR4A3 e3	5	3	5′-GGGCATATCTTCAATC-3′	37 → ≥3	≥2	≤0	0	238	193	3	11 (21)
TFG e7::NR4A3 e3	5	3	5′-GGGCATATCTGAATAC-3′	27 → ≥3	≥3	≤3	1	43	12	4	4 (26)

Table 3. The 9 designs with no sense-strand near-match at the default search depth. Six of these lose the property when the same junctions are re-screened at a tenfold deeper alignment ceiling, three of them having returned no near-match at all here; §3.10 reports that measurement and names the three that survive it. This table is the default-depth result, retained because it is the depth at which the corpus-wide counts elsewhere in the paper were computed. Every design at the 6 junctions where one exists. A design qualifies only if its retained hit list is not truncated — no more near-matches than the 15 the screens store — because the strand of an unstored hit cannot be recovered, so a truncated list cannot establish that nothing on the sense strand remains. The underlying search is itself capped, so these are the designs whose near-match lists are shortest, not the designs whose lists are known to be exhaustive. ΔΔG°37 is the margin by which the fusion duplex is favoured over the best duplex either parent can form, for an unmodified DNA:RNA hybrid; because the fusion duplex pairs both LNA wings and each parent duplex only one, it is a lower bound on the modified oligonucleotide's discrimination rather than an upper one. None of these numbers is a measurement of off-target activity, and none speaks to cleavage.

[3] Under the optimistic five-fold and the pessimistic no-discrimination bound on RNase-H1 single-mismatch discrimination. A single value means the two bounds agree.

[4] Of four conventional antisense design rules: GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, no CpG dinucleotide.

[5] Whether the design still carries no sense-strand near-match once its junction is re-screened at the tenfold deeper ceiling. The verdict is computed from the three deep columns beside it, not asserted, so this table cannot come to disagree with §3.10 about which designs survive. The six that do not are the reason this table's default-depth zeros must not be read on their own.

design	junction	GC (%)	gap-level margin	ΔΔG°37 (kcal/mol)	near-matches, either strand	of those, on the sense strand	exact / ≤1-mismatch matches	residual cleavage load, both bounds³	conventional rules failed⁴	at the deeper ceiling: near-matches	of those, on the sense strand	loci with a gap-spanning hit	survives⁵
5′-GCATATCCGTGGACGC-3′	EWSR1 e1::NR4A3 e3	62.5	1	7.981	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	84	83	1	**no**
5′-GGCATATCCGTGGACG-3′	EWSR1 e1::NR4A3 e3	62.5	2	10.085	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	29	22	0	**no**
5′-GGGCATATCCGTGGAC-3′	EWSR1 e1::NR4A3 e3	62.5	3	12.189	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	27	18	0	**no**
5′-AGGGCATATCGGAGTC-3′	FUS e8::NR4A3 e3	56.2	2	10.895	3	0	0 / 0	0	contains a CpG	3	0	0	yes
5′-GGGCATATCCGACATG-3′	TAF15 e1::NR4A3 e3	56.2	3	11.894	5	0	0 / 0	0	contains a CpG	5	0	0	yes
5′-GGGCATATCTCTATAA-3′	TCF12 e17::NR4A3 e3	37.5	3	8.556	8	0	0 / 0	0	GC outside 40–60%	118	101	5	**no**
5′-GCATATCAAGCGCTGC-3′	TCF12 e7::NR4A3 e3	56.2	1	7.98	1	0	0 / 0	0	contains a CpG	18	2	0	**no**
5′-GGCATATCAAGCGCTG-3′	TCF12 e7::NR4A3 e3	56.2	2	10.085	2	0	0 / 0	0	contains a CpG	2	0	0	yes
5′-CAGGGCATATCTTGCA-3′	TCF12 e9::NR4A3 e3	50.0	1	9.325	7	0	0 / 0	0	none	67	18	4	**no**

Table 4. The best available design at each of the 38 in-frame junctions. Tables 2 and 3 select across the panel; this table selects within each junction, which is the question a patient's fusion poses. Designs are ranked by parent liability first, since sparing the wild-type parents is what the modality exists for, then by pre-mRNA sites, then by distinct gene loci, with ties broken on gap-level margin rather than on raw hit counts. Nothing was re-screened: every field is joined from a screen already reported above. Whether a junction has a published exon-resolved breakpoint is reported separately from specificity and never folded into the ranking — “published” means an exon-resolved EMC breakpoint is reported for that exon pair, “exon not reported” that the partner has a resolved breakpoint at a different exon, and “none published” that no exon-resolved breakpoint has been reported for that partner at all. The last of those is absence of evidence: EMC case reports usually name the partner gene without sequencing to nucleotide resolution. Gap-paired near-matches are at the tenfold deeper alignment ceiling, where every hit list is complete. The genome column is the observed number of gap-paired sites at ≤2 mismatches over the number expected for an arbitrary 16-mer, so 1.00 is chance. A junction with no design clearing the parent screen is reported as such rather than given a best row.

junction	exon-resolved breakpoint	designs clearing the parent screen	best available design	gap-level margin	longest parent duplex through the gap (bp)	gap-paired near-matches at the deeper ceiling (transcripts → loci)	genome-wide gap-paired load, observed/expected
EWSR1 e10::NR4A3 e3	exon not reported	3 of 5	5′-GGCATATCTAGATCAA-3′	2	7	70 → 4	0.61
EWSR1 e12::NR4A3 e3	published	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
EWSR1 e13::NR4A3 e3	published	3 of 5	5′-GGGCATATCTCCACGG-3′	3	8	24 → 2	0.29
EWSR1 e15::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCCGGGGGC-3′	3	0	33 → 1	0.06
EWSR1 e1::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCCGTGGAC-3′	3	0	0 → 0	0.07
EWSR1 e4::NR4A3 e3	exon not reported	4 of 5	5′-AGGGCATATCAGTGGG-3′	2	6	9 → 3	0.71
EWSR1 e7::NR4A3 e3	exon not reported	5 of 5	5′-CAGGGCATATTCTGCT-3′	1	8	14 → 5	1.08
EWSR1 e9::NR4A3 e3	exon not reported	4 of 5	5′-GGCATATCACCAGGCT-3′	2	7	15 → 4	0.73
FUS e10::NR4A3 e3	none published	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
FUS e11::NR4A3 e3	none published	3 of 5	5′-GGGCATATCTCCTCGC-3′	3	9	30 → 1	0.22
FUS e13::NR4A3 e3	none published	4 of 5	5′-GGGCATATCCATGTGA-3′	3	7	2 → 2	0.75
FUS e1::NR4A3 e3	none published	2 of 5	5′-GGGCATATCGTTTGAG-3′	3	8	5 → 2	0.12
FUS e3::NR4A3 e3	none published	5 of 5	5′-GGCATATTGTTCTGGC-3′	2	7	18 → 3	0.76
FUS e5::NR4A3 e3	none published	2 of 3	5′-GGGCATATCTCCACCT-3′	3	8	33 → 4	0.65
FUS e7::NR4A3 e3	none published	4 of 5	5′-GGCATATCACCAAATT-3′	2	7	10 → 3	0.92
FUS e8::NR4A3 e3	none published	3 of 5	5′-AGGGCATATCGGAGTC-3′	2	0	0 → 0	0.06
TAF15 e11::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
TAF15 e12::NR4A3 e3	exon not reported	3 of 5	5′-GGGCATATCTCGCCGC-3′	3	6	4 → 2	0.04
TAF15 e14::NR4A3 e3	exon not reported	0 of 5	—	—	—	—	—

junction	exon-resolved breakpoint	designs clearing the parent screen	best available design	gap-level margin	longest parent duplex through the gap (bp)	gap-paired near-matches at the deeper ceiling (transcripts → loci)	genome-wide gap-paired load, observed/expected
TAF15 e1::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCCGACATG-3′	3	0	0 → 0	0.04
TAF15 e4::NR4A3 e3	exon not reported	2 of 5	5′-GGCATATCTGACTGAC-3′	2	8	12 → 2	0.69
TAF15 e6::NR4A3 e3	published	1 of 5	5′-GGGCATATCTTGTGTG-3′	3	9	8 → 5	0.60
TAF15 e8::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCACCAAAA-3′	3	7	36 → 3	0.86
TAF15 e9::NR4A3 e3	exon not reported	2 of 5	5′-AGGGCATATCAGCATC-3′	2	6	6 → 2	1.13
TCF12 e11::NR4A3 e3	exon not reported	3 of 5	5′-GGCATATCTAGAATGC-3′	2	8	4 → 3	0.69
TCF12 e13::NR4A3 e3	exon not reported	2 of 5	5′-GGCATATCTGTGAGAG-3′	2	7	6 → 5	1.08
TCF12 e17::NR4A3 e3	exon not reported	3 of 5	5′-GGGCATATCTCTATAA-3′	3	7	14 → 5	0.83
TCF12 e19::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCTCTGACT-3′	3	8	75 → 3	0.82
TCF12 e3::NR4A3 e3	exon not reported	0 of 5	—	—	—	—	—
TCF12 e5::NR4A3 e3	published	4 of 5	5′-GGGCATATCCATCAGA-3′	3	7	17 → 1	0.56
TCF12 e7::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCAAGCGCT-3′	3	8	0 → 0	0.26
TCF12 e9::NR4A3 e3	exon not reported	1 of 5	5′-GGGCATATCTTGCATA-3′	3	8	39 → 6	0.71
TFG e2::NR4A3 e3	none published	0 of 4	—	—	—	—	—
TFG e3::NR4A3 e3	none published	4 of 5	5′-GGGCATATCAAATAAT-3′	3	8	18 → 6	1.35
TFG e4::NR4A3 e3	none published	1 of 5	5′-AGGGCATATCATTTTC-3′	2	7	82 → 12	1.77
TFG e5::NR4A3 e3	none published	2 of 5	5′-GGCATATCTGAAACCT-3′	2	8	34 → 7	1.38
TFG e6::NR4A3 e3	none published	3 of 5	5′-GGGCATATCTTCAATC-3′	3	9	29 → 3	0.73
TFG e7::NR4A3 e3	published	2 of 5	5′-GGCATATCTGAATACT-3′	2	9	24 → 6	1.24

Table 5. Gap length against junction specificity, at one junction and across the design space. The same junctions tiled and screened at three gapmer geometries, wing held at five nucleotides so that only the catalytic gap changes. Inside the gap the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap, which the generating module asserts for every design rather than assuming. Within one geometry the gap is fixed, so each nucleotide of gap-level margin is one FEWER nucleotide of contiguous wild-type-parent duplex; what raises the parent-paired run at every register is a longer gap, which is the only way past a geometry's ceiling of half its gap rounded down. The two directions are reported separately and never combined into a score. Near-match counts fall partly for a reason the instrument guarantees rather than measures: at a fixed budget of two mismatches every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the set can only shrink, and two mismatches is a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall and which designs reach zero are measurements. The three blocks carry different denominators and are not comparable across blocks: the junction block is one molecule, the matched-junction block is the six junctions every geometry was screened at, and the corpus block is each geometry's whole design space, which is not screened at the same junctions. The exhaustive GRCh38 genome scan is unavailable at 18 and 20 nucleotides by construction, so no row reports it. Because the wing is five throughout, a parent's junction hybrid is five base pairs plus its share of the gap, so pairing five nucleotides of contiguous gap DNA and reaching a ten-base-pair hybrid are the same condition and are reported as one row. ΔG°37 values are for an unmodified DNA:RNA hybrid; the wing is five at every geometry, so LNA affinity enters each parent duplex identically and cannot explain a difference between the columns. None of these numbers is a measurement of cleavage.

	5-6-5 (16-mer)	5-8-5 (18-mer)	5-10-5 (20-mer)
At the *EWSR1* e12 / *TAF15* e11 / *FUS* e10 junction			
design (5′→3′)	GGGCATATCATCAAAC	AGGGCATATCATCAAACC	CAGGGCATATCATCAAACCA
gap-level margin	3	4	5
sense-strand gap-spanning cleavage risks	123	3	0
gene loci carrying one	6	1	0
near-matches (≤2 mismatches, deeper ceiling)	189	50	20
≤1-mismatch matches over 186,185 transcripts	1	0	0
mature-parent duplex through the whole gap (bp)	8 (*TFG*)	0	0
contiguous DNA a wild-type parent pairs (nt)	3	4	5
most stable parent ΔG°37 (kcal/mol)	−7.77	−8.66	−10.25
Over the six junctions screened at every geometry			
designs screened	30	42	54
median near-matches	86.5	15	0
median gap-spanning cleavage risks	21	0	0
designs carrying none	8 of 30	28 of 42	54 of 54
most risk loci on any one design	7	2	0
designs with no near-match at all	0 of 30	7 of 42	39 of 54
Over each geometry's whole design space			
junction-spanning registers per junction	5	7	9
fusion-specific designs	190	266	342
best gap-level margin available	3	4	5
a mature parent can pair the whole gap	181 of 190	130 of 266	87 of 342

	5-6-5 (16-mer)	5-8-5 (18-mer)	5-10-5 (20-mer)
parent pairs ≥5 nt of contiguous gap DNA, a ten-base-pair hybrid	76 of 190	228 of 266	342 of 342
designs pairing the gap in parent pre-mRNA	19 of 190	11 of 266	9 of 342
median most stable parent $\Delta G°37$ (kcal/mol)	−8.66	−11.60	−14.58

Table 6. Where the clinically-relevant reagents' off-target loci are expressed. Every gene locus returned by the deeper screens at four of the five junctions with a published exon-resolved EMC breakpoint, read against reference expression data. The two compartments answer different questions and are never combined: a systemically dosed phosphorothioate gapmer distributes predominantly to liver and kidney, so liver, kidney - cortex and kidney - medulla address exposure, while the soft-tissue column is the normal tissue of the compartment EMC arises in and stands in for a tumour no reference atlas contains. Values are GTEx v8 median TPM across each tissue's donors. The two cuts behind the last column are stated for legibility and are not thresholds of concern: below 1 TPM in all three exposure tissues reads as below detection, at or above 10 TPM in any of them as the level at which an off-target hypothesis would have to be tested. Every raw median is released so another cut can be applied without re-running. Tiling registers is how many of the designs tiled across that junction return the locus, which is robustness to where the window is placed and is a different axis from the record count beside it; neither is ranked on. Transcript records are how many accessions RefSeq lists for the gene, that is annotation depth, not expression and not affinity. A locus with no reading carries the reason rather than a zero, because an absent reading is not a reading of absence. Every hit behind this table sits at 14 of 16 identity, the loosest the screen admits, so nothing here distinguishes these loci from one another on affinity. None of these numbers is a measurement of cleavage, and no expression figure is a predicted cleavage event.

junction	gene locus	transcript records	tiling registers returning it	Liver	Kidney - Cortex	Kidney - Medulla	soft-tissue proxy maximum	exposure-organ reading
EWSR1 e12::NR4A3 e3	*ANKS1B*	67	1 of 1	0.03	0.46	0.28	3.6 (Artery - Tibial)	below the lower cut in all three
	ZNF667	37	1 of 1	0.31	1.63	2.58	6.2 (Nerve - Tibial)	detectable, below the upper cut
	GMCL1	9	1 of 1	4.52	4.98	6.72	18.3 (Artery - Tibial)	detectable, below the upper cut
	LOC105374140	5	1 of 1	—	—	—	—	no gene model — not measurable
	LOC105370997	4	1 of 1	—	—	—	—	no gene model — not measurable
	CHST5	1	1 of 1	0.07	0.35	0.78	0.8 (Nerve - Tibial)	below the lower cut in all three
TAF15 e6::NR4A3 e3	*G3BP2*	56	2 of 5	11.92	17.03	23.12	77.0 (Cells - Cultured fibroblasts)	at or above the upper cut
	LINC02030	22	2 of 5	0.00	0.00	0.02	0.3 (Skin - Sun Exposed (Lower leg))	below the lower cut in all three
	MIR9-2HG	18	2 of 5	—	—	—	—	no reading taken
	LAMA4	13	1 of 5	1.28	3.13	6.06	268.6 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	ZFPM2	12	3 of 5	0.64	0.55	0.40	9.6 (Artery - Tibial)	below the lower cut in all three
	GNAL	10	2 of 5	0.66	1.74	1.81	8.6 (Artery - Tibial)	detectable, below the upper cut
	NRP1	5	5 of 5	6.62	16.87	17.81	104.7 (Cells - Cultured fibroblasts)	at or above the upper cut
	SLC17A3	4	2 of 5	9.14	33.61	8.09	0.0 (Adipose - Subcutaneous)	at or above the upper cut
	CA5B	3	3 of 5	0.66	1.57	1.96	11.7 (Artery - Tibial)	detectable, below the upper cut
	CA5BP1-CA5B	3	3 of 5	—	—	—	—	no reading taken
	EEFSEC	3	1 of 5	16.15	14.37	15.57	39.1 (Nerve - Tibial)	at or above the upper cut
	ANKRD26P3	1	1 of 5	0.00	0.00	0.00	0.0 (Muscle - Skeletal)	below the lower cut in all three
	GBP4	1	1 of 5	3.12	5.59	10.08	18.2 (Adipose - Subcutaneous)	at or above the upper cut

junction	gene locus	transcript records	tiling registers returning it	Liver	Kidney - Cortex	Kidney - Medulla	soft-tissue proxy maximum	exposure-organ reading
	LOC105376349	1	1 of 5	—	—	—	—	no gene model — not measurable
	LOC124907518	1	1 of 5	—	—	—	—	no gene model — not measurable
	NRXN3-AS1	1	1 of 5	—	—	—	—	no reading taken
	ST3GAL1	1	1 of 5	28.58	16.36	8.06	27.2 (Muscle - Skeletal)	at or above the upper cut
EWSR1 e13::NR4A3 e3	*FNBP1*	42	2 of 5	7.08	12.12	14.06	56.5 (Nerve - Tibial)	at or above the upper cut
	EHMT2	34	2 of 5	6.89	13.23	16.70	53.0 (Nerve - Tibial)	at or above the upper cut
	ZNF215	27	1 of 5	0.13	0.67	1.27	5.2 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	ESYT2	26	2 of 5	10.42	14.26	28.28	139.9 (Artery - Tibial)	at or above the upper cut
	LRP5L	8	2 of 5	7.83	6.32	11.16	8.0 (Nerve - Tibial)	at or above the upper cut
	CDC42SE1	6	2 of 5	22.74	44.73	78.97	187.6 (Nerve - Tibial)	at or above the upper cut
	THEMIS	6	1 of 5	0.13	0.08	0.18	0.3 (Adipose - Subcutaneous)	below the lower cut in all three
	LOC105374651	4	1 of 5	—	—	—	—	no gene model — not measurable
	ZC3H4	4	1 of 5	6.41	5.89	9.71	26.2 (Nerve - Tibial)	detectable, below the upper cut
	ERBIN	2	1 of 5	9.80	9.38	14.68	43.8 (Cells - Cultured fibroblasts)	at or above the upper cut
	ZNF236	2	1 of 5	1.68	2.29	2.90	8.7 (Nerve - Tibial)	detectable, below the upper cut
TCF12 e5::NR4A3 e3	*HNRNPA2B1*	100	2 of 5	184.12	247.24	457.30	656.6 (Nerve - Tibial)	at or above the upper cut
	PIK3CG	51	3 of 5	0.14	0.17	0.37	1.2 (Adipose - Subcutaneous)	below the lower cut in all three
	MROH2A	28	1 of 5	1.04	0.42	0.07	0.3 (Skin - Sun Exposed (Lower leg))	detectable, below the upper cut
	LINC02940	9	1 of 5	—	—	—	—	no reading taken
	EXOC2	6	1 of 5	5.27	6.54	8.87	22.5 (Nerve - Tibial)	detectable, below the upper cut
	KCNG3	4	2 of 5	0.00	0.09	0.06	1.3 (Nerve - Tibial)	below the lower cut in all three
	PLAC8	4	2 of 5	1.43	0.32	0.30	0.7 (Adipose - Subcutaneous)	detectable, below the upper cut
	EFCAB11	2	1 of 5	0.37	0.92	1.20	2.1 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	LOC107984281	2	2 of 5	—	—	—	—	no gene model — not measurable
	LOC107985219	2	2 of 5	—	—	—	—	no gene model — not measurable
	LOC107987169	1	1 of 5	—	—	—	—	no gene model — not measurable
	LOC124905457	1	1 of 5	—	—	—	—	no gene model — not measurable

Figure legends

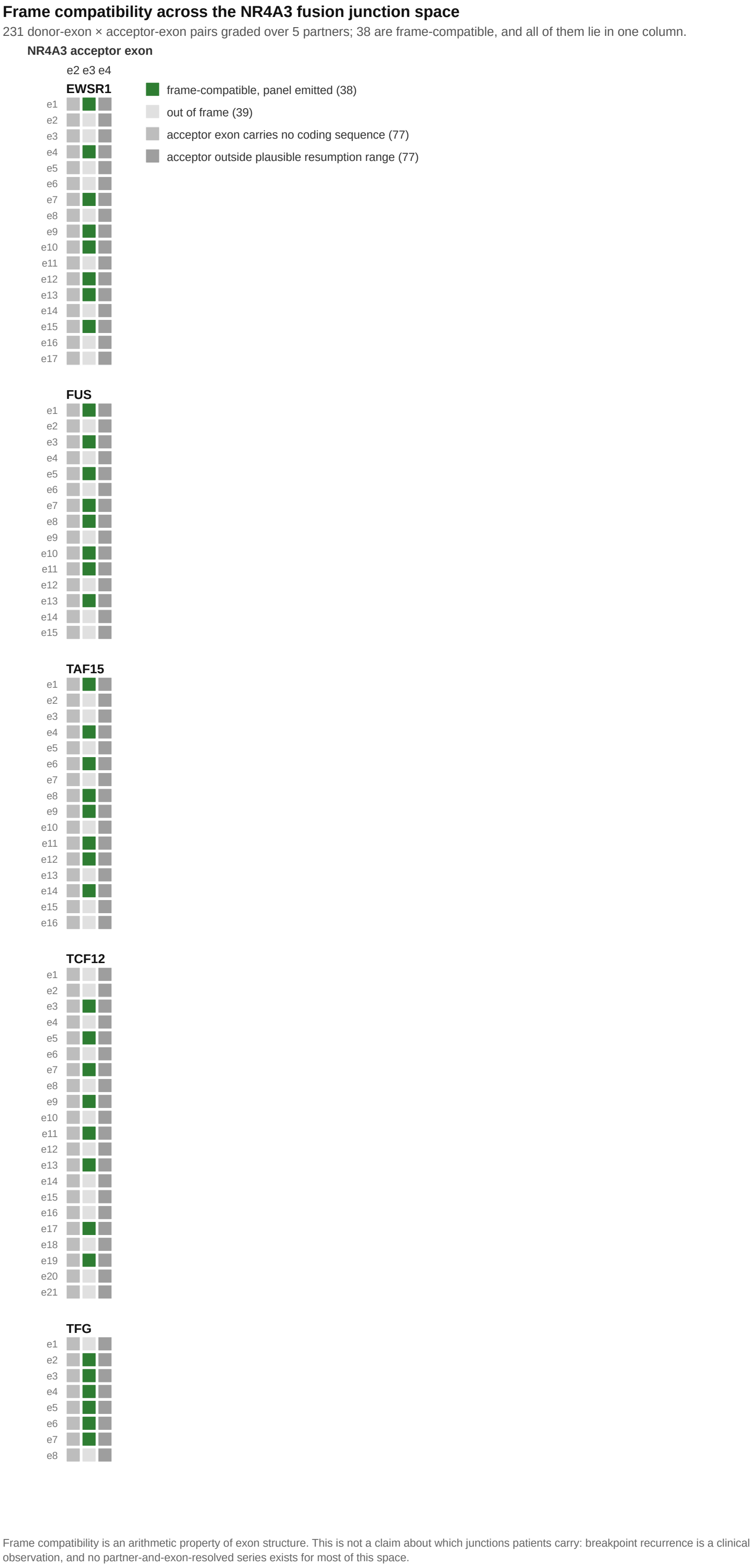


Figure 1. Reading-frame compatibility across the NR4A3 fusion junction space. All 231 donor-exon × acceptor-exon pairs across *EWSR1*, *TAF15*, *TCF12*, *FUS* and *TFG*, graded against the frame condition. Rows are donor exons grouped by partner; columns are *NR4A3* acceptor exons. Two acceptor columns are refused in every pair for structural reasons, so the 38 in-frame junctions lie in a single column.

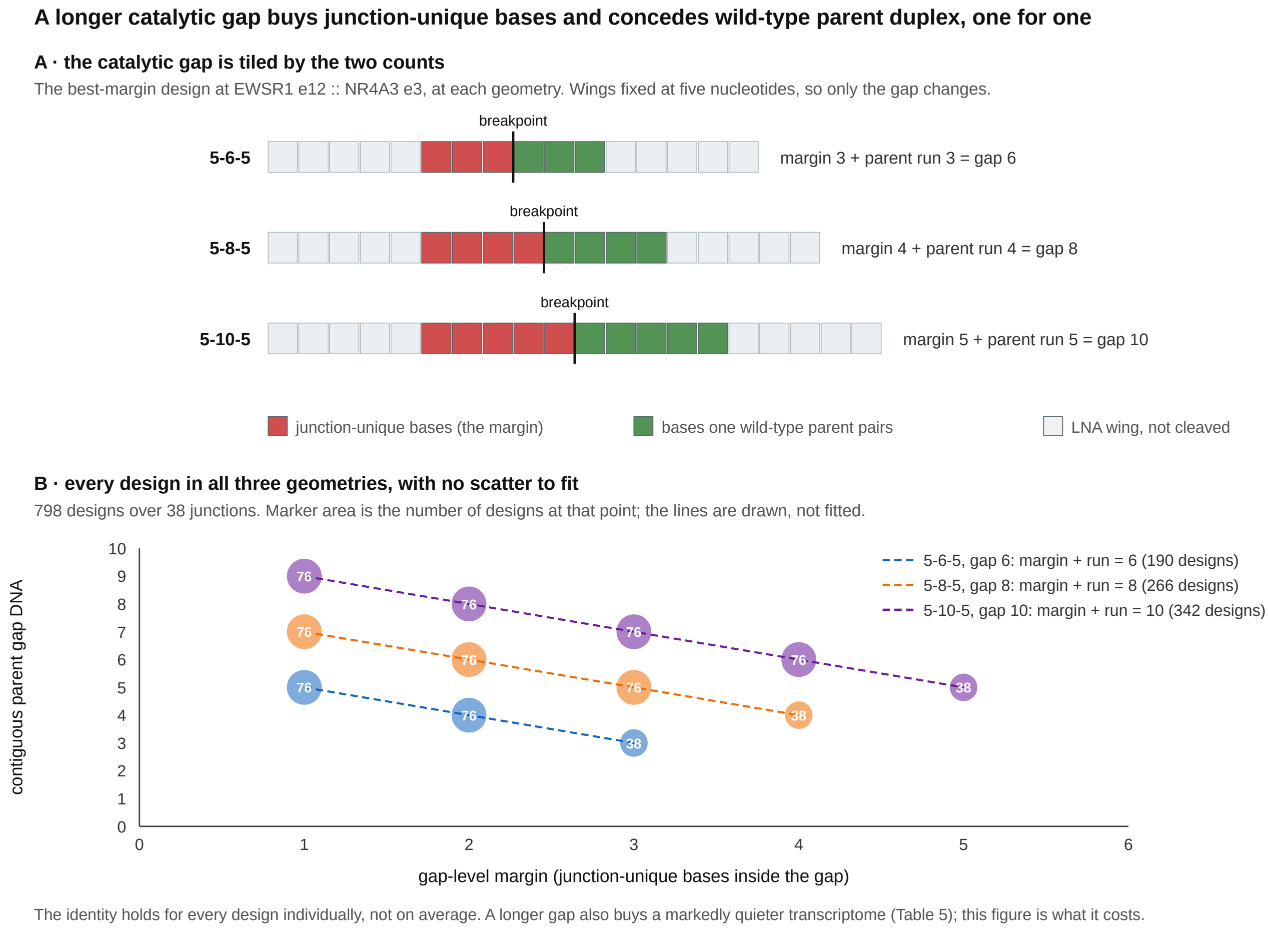


Figure 2. The margin a longer catalytic gap wins is the parent duplex it concedes. (A) The best-margin design at *EWSR1* exon 12 joined to *NR4A3* exon 3, drawn at 5-6-5, 5-8-5 and 5-10-5 with the wings held at five nucleotides. Every base inside the catalytic gap comes from the donor exon or from the acceptor exon, so the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side tile the gap and sum to it. (B) Every fusion-specific design in all three geometries, 798 over 38 junctions, plotted as gap-level margin against the contiguous run of gap DNA a wild-type parent can pair. Marker area is the number of designs at that point and the label is that count; the three lines are drawn from the identity, not fitted. The relation holds for each design individually rather than on average. Within one geometry the two move inversely along a line of slope −1, so a higher-margin register concedes a shorter parent duplex; what a geometry fixes is the ceiling on margin, at half its gap rounded down, and clearing that ceiling means a longer gap and a higher parent-paired run at every register. A longer gap also buys a markedly quieter transcriptome (§3.8, Table 5); this figure is what it costs.

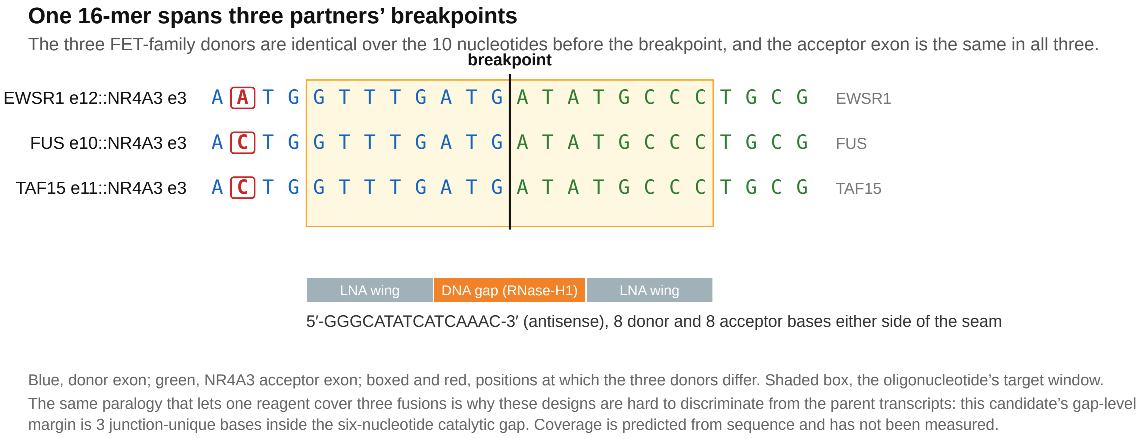
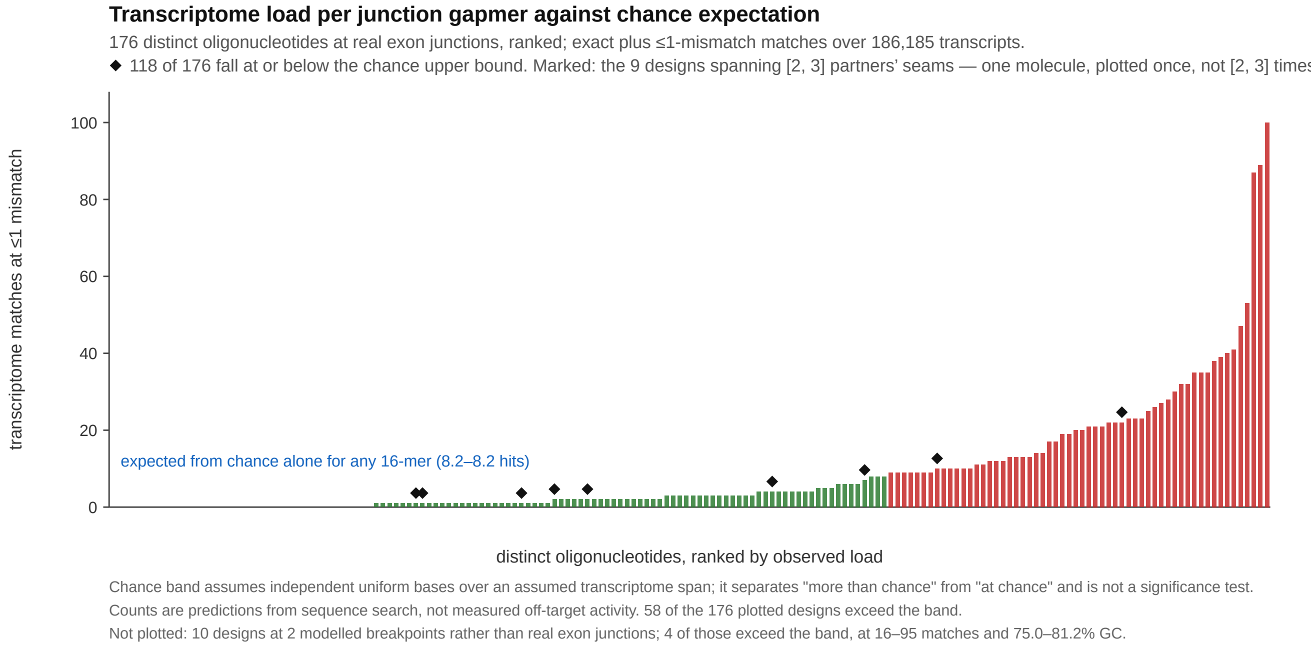


Figure 3. One 16-mer spans three partners' breakpoints. The junction windows of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, aligned at the breakpoint. Blue, donor exon; green, acceptor exon; positions at which the three donors differ are boxed as well as coloured, for greyscale and colour-blind readers. The shaded box is the target window of 5′-GGGCATATCATCAAAC-3′, with the 5-6-5 gapmer architecture below it and its gap-level margin of three alongside. The three donors are identical over the ten nucleotides before the breakpoint, which is what makes one oligonucleotide junction-spanning at all three junctions. Coverage is predicted from sequence and has not been measured.



Supplementary Figure S1. Transcriptome load per design against chance expectation. Each bar is one distinct oligonucleotide's count of exact plus ≤1-mismatch matches over 186,185 transcripts, ranked. The 190 design records at real exon junctions collapse to 176 molecules, because nine of the 16-mers are junction-spanning at more than one partner's junction at once — five at three junctions and four at two — and each of those is one physical oligonucleotide, plotted once rather than repeatedly (marked). The line is the number of such matches expected for an arbitrary 16-mer under an independent-uniform-base null, 8.2, computed against the scan's measured 718,571,139-nucleotide span; 118 of the 176 fall at or below it and 58 exceed it. Ten further designs from modelled breakpoints not built from a spliced transcript model are excluded, and are released with the artefacts. It is an expected count: the observed mean is 9.2, a ratio of 1.12, while the median is 3, so real transcript sequence produces a long right tail the null cannot rather than a uniform shift away from it. The line separates "more than chance" from "at chance" and is not a significance test; the counts are predictions from sequence search, not measured off-target activity.

Declarations

Data and code availability. [ARCHIVE DOI], deposited from `github.com/trimcrae/Rare-cancers`. A manifest listing every archived file with its SHA-256 travels with the deposit. Artefacts include the graded junction atlas, per-junction design panels, all five screens, the per-junction reagent table behind Table 4, the graded re-scores under both discrimination bounds, and the retrieval records for every literature claim.

Provenance and corrections. An earlier version of these analyses placed the acceptor junction incorrectly through a coding-versus-transcript exon indexing error and was withdrawn in full; all panels were rebuilt and verified against two independent transcript acquisitions. The complete correction record, including every superseded value, is released with the archive.

Because that is the failure a reader will reasonably assume could recur, the two instruments the paper's conclusions rest on were reimplemented a second time and the two implementations compared. The reimplementation shares no code with the original and differs from it on four axes: it splices each mature transcript out of the genomic record rather than reading the cDNA record; it locates each gene's coding start by open-reading-frame search rather than by reading the annotated 5′ untranslated length, which is the class of value the retracted error turned on; it grades the reading frame by arithmetic on exon coding-length vectors rather than by translating the chimera; and it computes the mature-parent screen by substring search over the design's gap-containing substrings rather than by scanning every parent window and extending outward from the gap. The two agree on all 231 graded exon pairs, field by field and not only on the grade, and on the longest parent duplex of all 190 designs, giving the same 87 and the same 61 against *NR4A3*. The two transcript acquisitions agree base for base for all six genes, and the annotation-free coding start reproduces the annotated one for all six. Both implementations, the comparison and its deliberate-corruption tests are in the archive.

Two things that check should not be read as. It is not external review: the same author prepared both implementations. And it bounds implementation error only — two implementations of a specification that is itself wrong will agree with each other and both be wrong, so agreement here is not evidence that the longest contiguous run containing the catalytic gap is the right quantity to compute. Independent review of the code by another group remains wanted and is not claimed.

Competing interests. The author declares no financial competing interests: he holds no position, equity, consultancy or patent relating to any gene, sequence or agent named here, and no oligonucleotide described in this manuscript has been synthesised, licensed or offered for sale. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Funding. This work received no external funding and was self-funded by the author. No grant, institution, company or charity supported it, and no funder had any role in the design of the analyses, the interpretation of the results, or the decision to publish.

Ethics. No human subjects, human material or animals were involved. All clinical figures are taken from published reports and are cited.

Use of AI tools. A large language model (Claude, Anthropic) was used throughout this work: to write the analysis code, to run the graded design and screening pipelines, to draft and revise this manuscript, and to conduct internal critical review of earlier drafts. Every quantitative statement derived from sequence or from a screen is produced by code in the released archive and is reproducible from it, while the clinical figures are transcribed from the publications cited for them; no numerical result was generated by a language model directly. Every literature identifier was checked against a retrieved bibliographic record, and identifiers that could not be so anchored were removed rather than retained. The three references dated 2026 were additionally checked for live resolution against two independent registries, Europe PMC and Crossref, which return the same title and the same digital object identifier for each; those records travel with the archive. The frame grading and the mature-parent screen were additionally reimplemented and cross-checked as described under Provenance, which bounds implementation error but is not independent review. The author takes full responsibility for the content, including for the correctness of the code and for the interpretation of the results.

References

52 entries, numbered by first citation in the submission manuscript. Metadata is read from retrieved bibliographic records; an unretrieved field is left absent rather than completed.

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